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A survey of fake news detection: A comprehensive review of multimodal approaches

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ABSTRACT

With the rapid advancement of social media and artificial intelligence, fake news emerges as a critical societal challenge, contributing to widespread fake news and societal polarization. To address these challenges, multimodal approaches are considered promising solutions that integrate diverse data modalities to capture complementary information. This study provides a comprehensive examination of the evolution of fake news detection models, emphasizing the distinctive potential of multimodal techniques. We propose a novel taxonomy for categorizing multimodal approaches, which is structured around three critical dimensions: feature extraction, fusion methods, and learning objectives. Our review begins with a detailed comparison of key datasets, highlighting their characteristics, limitations, and applicability to multimodal tasks. We then conduct an in-depth analysis of multimodal detection techniques, exploring their methodologies and performance. In addition, we explore emerging technologies such as generative AI for robust detection and possible ethical considerations, including privacy risks and model biases. To address challenges such as cross-platform generalizability and real-world scalability, we propose several future directions and opportunities for developing effective multimodal fake news detection systems. Through this comprehensive review, we aim to contribute to the ongoing efforts to combat fake news by providing valuable insights and actionable guidance for researchers and practitioners in the field.

1. Introduction

In today's digital society, the rapid propagation of fake news has become a critical societal challenge that undermines public trust and exacerbates political polarization. The ubiquity of social media platforms, which a majority of Americans now rely on for news [1], has amplified the creation and spread of false information, making it increasingly difficult for individuals to distinguish between real and fake content [2,3].

The societal impacts of fake news are multifaceted, affecting critical domains such as politics [4], public health [5] and economic stability [6]. For instance, during the 2016 U.S. presidential election, numerous fake news stories circulated across social media platforms, with studies indicating their influence on voters' electoral decisions [7]. Beyond politics, fake news poses significant risks

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虚假新闻检测综述：多模态方法的全面回顾

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述

ABSTRACT

随着社交媒体和人工智能的快速发展, 假新闻成为一项关键的社会挑战, 加剧了虚假新闻的泛滥和社会极化。为了应对这些挑战, 多模态方法被认为是极具前景的解决方案, 它们通过整合多样化的数据模态来捕捉互补信息。本研究全面审视了假新闻检测模型的演变, 强调了多模态技术的独特潜力。我们提出了一种新颖的多模态方法分类体系, 该体系围绕三个关键维度构建: 特征提取、融合方法和学习目标。我们的综述首先对关键数据集进行了详细比较, 强调了它们的特征、局限性以及对多模态任务的适用性。随后, 我们对多模态检测技术进行了深入分析, 探讨了它们的方法论和性能表现。此外, 我们还探讨了诸如生成式AI等用于鲁棒检测的新兴技术, 以及可能的伦理考量, 包括隐私风险和模型偏见。为了应对跨平台泛化性和现实世界可扩展性等挑战, 我们提出了未来开发有效多模态假新闻检测系统的几个方向和机遇。通过这篇全面的综述, 我们旨在为该领域的研究人员和从业者提供有价值的见解和可行的指导, 从而为打击假新闻的持续努力做出贡献。

1. 引言

在当今的数字社会中, 假新闻的快速传播已成为一项严峻的社会挑战, 它破坏了公众信任并加剧了政治极化。社交媒体平台的普及——如今大多数美国人都依赖其获取新闻[1], ——放大了虚假信息的制造与传播, 使得个人越来越难以区分真实与虚假内容[2,3]。

假新闻的社会影响是多方面的, 影响着政治[4], 公共卫生[5] 和经济稳定[6]等关键领域。例如, 在2016年美国总统选举期间, 大量假新闻在社交媒体平台上传播, 研究表明它们影响了选民的选举决定[7]。除了政治之外, 假新闻还构成了重大风险

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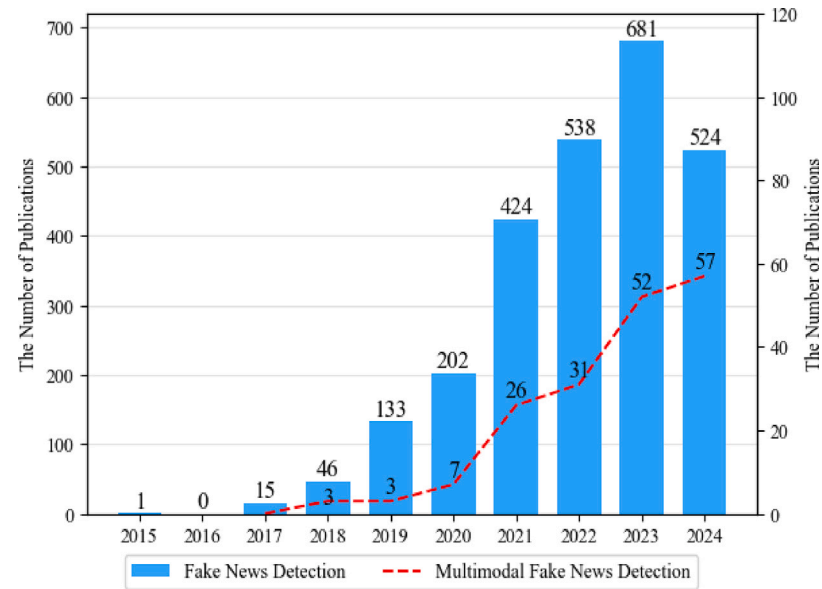


Fig. 1. Trend of fake news detection research from 2015 to 2024.

to public health. The COVID-19 pandemic was accompanied by an “infodemic” of false information regarding the virus and vaccine safety, which eroded public confidence in evidence-based health policies [8,9]. These examples underscore the profound societal threat that fake news poses across diverse sectors.

The proliferation of fake news emerged as a critical challenge, driving significant advancements and a growing body of research in fake news detection, as Fig. 1 shows the number of publications for fake news detection from 2015 to 2024. Early efforts in this domain primarily focused on analyzing single modalities, such as textual content [10] or visual elements [11,12] in isolation. However, such unimodal approaches are inherently limited, as they often fail to capture the complex interplay and inconsistencies between different data types that bad actors exploit to make fake news more persuasive and deceptive [13]. This limitation is becoming increasingly critical as modern social media platforms transition from text- and image-based content to video and audio formats [14,15].

In response to these limitations, multimodal approaches have emerged as robust solutions. Pioneering studies began integrating diverse data modalities, such as text and images, to capture complementary information and detect inconsistencies between them [16]. Subsequent advances in deep learning led to the development of powerful models, such as VisualBERT [17] and UNITER [18], which learn the correlations between textual and visual data to enhance detection accuracy. In addition to text and images, several researchers expanded to other modalities, such as audio [19] and social interaction information [20]. Studies have consistently demonstrated that multimodal frameworks outperform single-modality models. Consequently, research on multimodal fake news detection has rapidly expanded, a trend clearly illustrated by the growing number of publications in recent years, as shown in Fig. 1.

However, this rapid expansion led to a fragmented research landscape. Despite the proliferation of models, the field remains in its infancy, with notable research gaps and technical challenges that must be addressed. For instance, there is a significant lack of consensus on the relative advantages of various fusion strategies and optimal conditions for their application. Furthermore, many available datasets are limited to specific domains or languages, posing challenges in validating model performance in diverse real-world environments. Therefore, a systematic review is essential to organize this emerging body of work, identify key trends and limitations, and establish clear directions for future research.

Thus, this study aims to provide a comprehensive and structured analysis of the field by systematically examining key trends to guide future research. To achieve this, we introduce a novel taxonomy for categorizing multimodal approaches, which is structured around three critical dimensions: feature extraction, fusion methods, and learning objectives. Through a detailed comparison of key datasets and by applying our framework, we trace the development of detection models, provide a clear understanding of their underlying methodologies and critically evaluate their limitations. The goal is to offer valuable insights and actionable guidance for researchers and practitioners, ultimately contributing to the development of more robust and effective fake news detection systems.

The remainder of this paper is organized as follows. Section 2 reviews the prior literature and related surveys, identifying their limitations to establish the unique contribution of our comprehensive review. Section 3 provides a detailed analysis of the key datasets used in multimodal fake news detection, highlighting their attributes and limitations. Section 4 applies our proposed triaxial taxonomy to analyze and benchmark detection models across feature extraction, fusion strategies, and learning objectives. Finally, Section 5 discusses emerging challenges, such as AI-generated content and cross-platform generalizability, and outlines potential future research directions to advance the development of effective and scalable fake news detection systems.

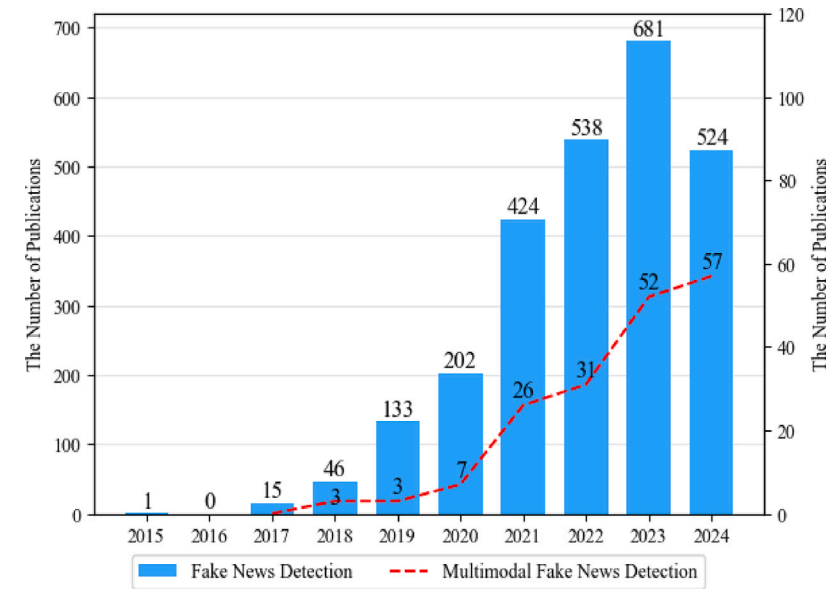


图 1. 2015 至 2024 年假新闻检测研究趋势。

对公共卫生。COVID-19大流行伴随着关于病毒和疫苗安全性的假信息的“信息疫情”，这削弱了公众对循证卫生政策的信心[8,9]。这些例子凸显了假新闻在各个领域构成的深刻社会威胁。

假新闻的泛滥已成为一项严峻挑战，推动了假新闻检测领域的重大进展和日益增多的研究，如图1所示，2015年至2024年间假新闻检测的出版物数量不断增加。该领域的早期工作主要集中于分析单一模态，例如孤立地分析文本内容[10]或视觉元素[11,12]。然而，这种单模态方法存在固有的局限性，因为它们往往无法捕捉不良行为者利用不同数据类型之间复杂的相互作用和不一致性，从而使假新闻更具说服力和欺骗性[13]。随着现代社交媒体平台从基于文本和图像的内容向视频和音频格式[14,15]转变，这一局限性正变得日益关键。

针对这些局限性，多模态方法应运而生，成为强有力的解决方案。开创性研究开始整合不同的数据模态，例如文本与图像，以捕捉互补信息并检测它们之间的一致性[16]。随后深度学习的进展推动了强大模型的开发，例如VisualBERT[17]和UNITER[18]，它们学习文本和视觉数据之间的相关性以提高检测准确率。除了文本与图像，一些研究人员还扩展到了其他模态，例如音频[19]和社交互动信息[20]。研究一致表明，多模态框架优于单模态模型。因此，多模态假新闻检测的研究迅速扩展，如图1所示，近年来出版物数量的增加清楚地说明了这一趋势。

然而，这种快速扩张导致了研究领域的碎片化。尽管模型激增，该领域仍处于起步阶段，存在必须解决的显著研究空白和技术挑战。例如，对于各种融合策略的相对优势及其应用的最佳条件，仍明显缺乏共识。此外，许多可用的数据集仅限于特定领域或语言，这对在多样化的真实世界环境中验证模型性能提出了挑战。因此，一项系统综述对于梳理这一新兴研究体系、识别关键趋势与局限性，并为未来研究确立明确方向至关重要。

因此，本研究旨在通过系统性地审视关键趋势以指导未来研究，从而对该领域提供全面且结构化的分析。为实现这一目标，我们引入了一种用于对多模态方法进行分类的新颖分类法，该分类法围绕三个关键维度构建：特征提取、融合方法和学习目标。通过对关键数据集的详细比较并应用我们的框架，我们追溯了检测模型的发展，清晰地理解了其底层方法，并批判性地评估了其局限性。目标是为研究人员和从业者提供有价值的见解和可操作的指导，最终促进更稳健、更有效的假新闻检测系统的发展。

本文的其余部分组织如下。第2节回顾了先前的文献和相关综述，指出了它们的局限性，从而确立了我们这篇全面综述的独特贡献。第3节对多模态假新闻检测中使用的关键数据集进行了详细分析，重点介绍了它们的属性和局限性。第4节应用我们提出的三轴分类法，从特征提取、融合策略和学习目标三个方面对检测模型进行分析和基准评估。最后，第5节讨论了新兴挑战，例如AI生成内容和跨平台泛化性，并概述了未来的潜在研究方向，以推动有效且可扩展的假新闻检测系统的发展。

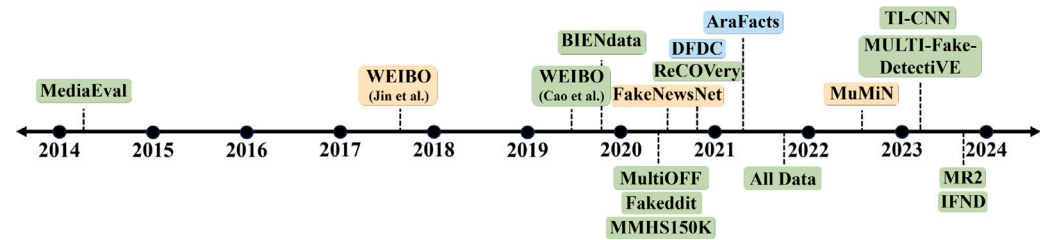


Fig. 2. Trends of multimodal fake news datasets; Green: Text + Image, Orange: Text + Image + social context metadata; Blue: Video. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

2. Related works

Early studies on fake news detection naturally gravitated toward written text. Shu et al. [21] and the comprehensive synthesis of Zhou and Zafarani [22] established a foundational taxonomy identifying four primary dimensions for deception detection: stylistic features, knowledge-based cues, propagation patterns, and source credibility assessment. During this formative period, empirical investigations were constrained by limited data availability, with researchers predominantly utilizing small-scale monolingual English datasets such as LIAR [23]. These studies primarily employed support vector machines, early convolutional neural networks (CNNs), and long short-term memory (LSTM) classifiers.

Once larger datasets and transformer models [24] appeared, attention shifted to resources and end-to-end architectures. D’ulizia et al. [25] cataloged more than forty corpora, documenting how they were built and where bias might lurk, while Alam et al. [26] listed dozens of CNN, Bi-LSTM, and BERT variants. Although these reviews are undoubtedly valuable, they offer only brief references to multimodal research, entirely overlooking the audio and video modalities. Moreover, they fail to consider how the structure and composition of a dataset may constrain the types of fusion strategies that a model can pursue.

Recently, the conversation has shifted again, firmly toward multimodality. Nasser et al. [27] followed a Precise Surveys and Meta-Examinations (PRISMA) style pipeline, screening almost a thousand records and retaining 121 studies that span text, image, video, and audio. Zidan et al. [28] focused on text-image fusion and compared transformer cross-attention and multimodal GNNs while raising questions about interpretability.

Early efforts conceptualized fake news detection almost exclusively as a text-classification task. They clustered algorithms around lexical, stylistic, and psycholinguistic cues under a single-modality assumption and relied predominantly on English, politics-oriented corpora. Consequently, the interaction between textual evidence and social context signals, such as user graphs, remains largely unexamined.

The proliferation of image-rich platforms motivated a next survey waves that appended brief “vision” subsections to traditional text-centric methods. Most of these studies analyzed a simplistic early- and late-fusion, omitted audio and video modalities, and offered only superficial treatment of higher-order operators such as cross-modal attention or bilinear pooling.

Building on these partial attempts, a more comprehensive strand of multimodal surveys appeared. Recent multimodal surveys were conducted to cover text-image, text-video, and text-audio research. However, these surveys still evaluate models separately and mostly ignore issues such as training data biases, metadata, speech signals, and advanced fusion components.

To address these gaps, we advance the field in an integrated manner.

- We introduce a unified tri-axial map that situates all studies by feature extractions, fusion strategies, and learning objectives
- We examine numerous widely used fake news datasets, expose modality and language biases, and suggest guidelines for curating balanced multimodal benchmarks
- We explore how emerging generative models reshape detection challenges and raise privacy- and bias-related considerations

3. Datasets

Fake news detection emerges as a critical research domain in modern society, aiming to assess the authenticity of information. Fake news detection models developed to address this issue rely heavily on trustworthy datasets for training and evaluation, and the diversity and quality of these datasets significantly influence the outcomes of research. Because fake news can propagate through various modalities, such as text, images, and videos, the importance of datasets incorporating multimodal data is becoming increasingly prominent.

Fig. 2 illustrates the growth trend of multimodal datasets, encompassing text, image, and video modalities. The emergence of multimodal datasets began in 2014, with their gradual adoption and development in the early years. However, a significant surge in the creation of multimodal fake news datasets has been observed since 2020, coinciding with rapid advancements in deep and multimodal learning research. This period marks a pivotal shift in the exploration and application of multimodal approaches, particularly in fields such as fake news detection. Multimodal datasets play a critical role in enhancing the accuracy and robustness of detection systems, making them indispensable for addressing the complexities of misinformation in various contexts.

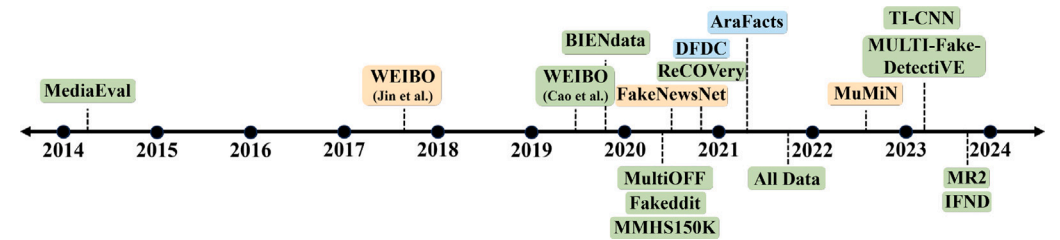


图2. 多模态假新闻数据集的趋势；绿色：文本 + 图像，橙色：文本 + 图像 + 社交上下文元数据；蓝色：视频。（关于本图例中颜色的解释，读者可参考本文的网络版。）

2. 相关 工作

早期关于假新闻检测的研究自然而然地倾向于书面文本。Shu等人[21] 以及Zhou和Zafarani的全面综合研究[22] 建立了一个基础分类法，确定了欺骗检测的四个主要维度：文体特征、基于知识的线索、传播模式和来源可信度评估。在这个形成期，实证研究受到数据可用性有限的制约，研究人员主要使用小规模的单语英语数据集，例如LIAR[23]。这些研究主要采用支持向量机、早期的卷积神经网络（CNN）和长短期记忆（LSTM）分类器。

随着更大型数据集和Transformer模型[24] 的出现，研究重点转向了资源和端到端架构。D’ulizia等人[25] 编目了四十多个语料库，记录了它们的构建方式以及可能存在偏见的环节，而Alam等人[26] 列出了数十种CNN、Bi-LSTM和BERT变体。尽管这些综述无疑具有价值，但它们仅对多模态研究作了简要提及，完全忽略了音频和视频模态。此外，它们未能考虑数据集的结构和组成如何制约模型可采用的融合策略。

最近，讨论的焦点再次转移，坚定地转向了多模态。Nasser等人[27] 遵循精确调查与荟萃分析（PRISMA）流程，筛选了近千条记录，并保留了涵盖文本、图像、视频和音频的121项研究。Zidan等人[28] 专注于文本-图像融合，比较了Transformer交叉注意力和多模态图神经网络，同时对可解释性提出了疑问。

早期的研究几乎将假新闻检测完全概念化为文本分类任务。他们在单模态假设下将算法围绕词汇、风格和心理语言学线索进行聚类，并主要依赖于英语的政治导向语料库。因此，文本证据与社交上下文信号（如用户图）之间的交互在很大程度上仍未被研究。

图像丰富平台的普及推动了下一波综述浪潮，这些综述在传统的以文本为中心的方法中附加了简短的“视觉”小节。这些研究大多分析了简单的早期和晚期融合，忽略了音频和视频模态，并且仅对跨模态注意力或双线性池化等高阶算子进行了肤浅的处理。

在这些局部尝试的基础上，出现了一系列更全面的多模态综述。最近的多模态综述涵盖了文本-图像、文本-视频和文本-音频研究。然而，这些综述仍然单独评估模型，并且大多忽略了训练数据偏差、元数据、语音信号和高级融合组件等问题。

为了解决 这些 差距，我们 以 一种 综合的 方式 推进 该 领域。

- 我们引入了一个统一的三轴图，通过特征提取、融合策略和学习目标来定位所有研究
- 我们考察了众多广泛使用的假新闻数据集，揭示了模态和语言偏差，并为构建平衡的多模态基准提出了指导原则
- 我们探讨了新兴的生成式模型如何重塑检测挑战，并引发了隐私和偏差相关考量

3. 数据集

假新闻检测作为现代社会中一个关键的研究领域应运而生，旨在评估信息的真实性。为解决这一问题而开发的假新闻检测模型严重依赖于值得信赖的数据集进行训练和评估，而这些数据集的多样性和质量显著影响着研究成果。由于假新闻可以通过各种模态传播，例如文本、图像和视频，因此包含多模态数据的数据集的重要性正变得日益突出。

图2展示了多模态数据集的增长趋势，涵盖文本、图像和视频模态。多模态数据集的出现始于2014年，并在早期逐渐被采用和发展。然而，自2020年以来，多模态假新闻数据集的创建出现了显著激增，这与深度和多模态学习研究的快速发展相吻合。这一时期标志着多模态方法的探索和应用发生了关键性转变，特别是在假新闻检测等领域。多模态数据集在提高检测系统的准确率和鲁棒性方面发挥着关键作用，使其成为应对各种上下文中错误信息复杂性的不可或缺的工具。



Fig. 3. Examples of fake news benchmark dataset.

Fig. 3 shows examples of fake news articles and their corresponding image samples. Datasets for fake news detection can be broadly categorized into unimodal and multimodal datasets. Each type captures the unique characteristics of a specific modality while providing the diverse metadata required to detect and understand the spread of fake news. This section systematically categorizes the commonly used datasets for fake news detection based on modality and discusses their key features and applications.

3.1. Unimodal-based dataset

Unimodal datasets for fake news detection focus on a single modality, such as text or image. These datasets play a foundational role in understanding fake news characteristics. A summary of the representative unimodal datasets is presented in Table 1.

Text-based datasets, the most widely used, typically include news articles, social media posts, and headlines labeled as fake or real. They provide insights into the linguistic patterns, sentiments, and grammatical features commonly associated with fake news.

For example, the LIAR [23] dataset serves as a benchmark for evaluating the veracity of political statements. It includes 12,836 statements collected from PolitiFact² between 2007 and 2016, each annotated with one of six fine-grained labels: pants-fire, false, barely-true, half-true, mostly-true, and true. These categories make the LIAR dataset particularly well-suited for assessing text-based fake news detection models that require a fine-grained classification. In addition to the headlines and bodies of news articles, the dataset contains metadata such as the source of the statement, contextual information, and the profession of the person making the statement. This rich contextual data enables researchers to examine fake news within a broader sociopolitical context. The LIAR dataset is primarily used for detecting political fake news and supports multi-class classification tasks, offering a valuable resource for advancing text-based fake news detection methods.

The ISOT [31] dataset is another important resource for political fake news detection. It consists of 12,600 real news articles from Reuters and 12,600 fake news articles from Kaggle. The fake news samples were characterized by sensationalist headlines and misleading content, making the dataset ideal for studying the linguistic patterns and emotional cues that differentiate fake news from legitimate news.

The CREDBANK [32] dataset is a large-scale text-based resource designed for credibility analysis research. It contains 60 million global tweets collected between October 2014 and February 2015, spanning 1000 events. Each event is annotated with a credibility rating, allowing researchers to evaluate the reliability of the information shared on social media platforms.

The FacebookHoax [33] dataset focuses on scientific and conspiracy-related news collected from Facebook, distinguishing between fake and real news stories. It includes 8923 hoax posts and 6577 non-hoax posts, making it a valuable resource for analyzing the spread and characteristics of fake news in social media ecosystems. This dataset enables research into the linguistic and contextual features that differentiate hoaxes from legitimate content, laying the groundwork for developing models to mitigate the impact of fake news on social media.

The PHEME [41] dataset supports rumor detection research by providing Twitter data related to major social events. It includes tweets collected from five significant incidents: the Ferguson unrest (August 2014), Ottawa shooting (October 2014), Sydney hostage crisis (December 2014), Charlie Hebdo attack (January 2015), and Germanwings plane crash (March 2015). The dataset is categorized into rumors and non-rumors, containing 1972 rumor tweets and 3830 non-rumor tweets. Each tweet is also classified according to the specific event it references, making the dataset particularly valuable for analyzing the propagation and contextual characteristics of rumors.

² <https://www.politifact.com/>.

图3展示了假新闻文章及其对应图像样本的示例。用于假新闻检测的数据集可大致分为单模态数据集和多模态数据集。每种类型都捕捉了特定模态的独特特征，同时提供了检测和理解假新闻传播所需的多样化元数据。本节根据模态系统地对假新闻检测的常用数据集进行分类，并讨论了它们的主要特征和应用。

3.1. 基于单模态的 数据集

单模态 数据集 用于 虚假 例如文本 或 图像。 这些 数据集 在理解虚假 新闻 特征方面 发挥着 基础 作用 al。 代表性 单模态 数据集 的 汇总 见 表 1。

基于文本的 数据集 是 使用最广泛的， 通常包含 新闻 文章、 社交 媒体帖子 以及 被标记为虚假 或 真实的 标题 真实。 这些 y p 提供 深入 g 了解， 了解 Lin gguistic p patterns， 情感， 以及 g 语法 特征 常见 y 与 假 新闻 相关。

例如， LIAR [23] 数据集可作为评估政治声明真实性的基准。它包含从PolitiFact² 2007年至2016年间收集的12,836条声明，每条声明都标注了六种细粒度标签之一：裤子着火、假、勉强真实、半真、大部分真和真。这些类别使得LIAR数据集特别适合评估需要细粒度分类的基于文本的假新闻检测模型。除了新闻文章的标题和正文外，该数据集还包含元数据，如声明来源、上下文信息以及发表声明者的职业。这些丰富的上下文数据使研究人员能够在更广泛的社会政治背景下考察假新闻。LIAR数据集主要用于检测政治假新闻，并支持多类分类任务，为推动基于文本的假新闻检测方法提供了宝贵的资源。

ISOT [31] 数据集是政治假新闻检测的另一个重要资源。它来自路透社的12,600篇真实新闻文章和来自Kaggle的12,600篇虚假新闻文章组成。虚假新闻样本以煽情标题和误导性内容为特征，使得该数据集非常适合研究区分假新闻与合法新闻的语言模式和情感线索。

CREDBANK [32] 数据集是一个大规模的文本资源，专为可信度分析研究而设计。它包含2014年10月至2015年2月期间收集的6000万条全球推文，涵盖1000个事件。每个事件都标注了可信度评级，使研究人员能够评估社交媒体平台上共享信息的可靠性。

FacebookHoax [33] 数据集关注从Facebook收集的科学新闻和阴谋论相关新闻，区分了虚假和真实新闻报道。它包含8923个恶作剧帖子和6577个非恶作剧帖子，使其成为分析假新闻在社交媒体生态系统中传播和特征的重要资源。该数据集能够研究区分恶作剧与合法内容的语言和上下文特征，为开发减轻假新闻对社交媒体影响的模型奠定了基础。

PHEME [41] 数据集通过提供与重大社会事件相关的Twitter数据来支持谣言检测研究。它包含了从五起重大事件中收集的推文：弗格森骚乱（2014年8月）、渥太华枪击案（2014年10月）、悉尼人质危机（2014年12月）、查理周刊袭击案（2015年1月）和德国之翼航空坠机事件（2015年3月）。该数据集分为谣言和非谣言，包含1972条谣言推文和3830条非谣言推文。每条推文还根据其引用的具体事件进行分类，使得该数据集在分析谣言的传播和上下文特征方面具有特别的价值。

² <https://www.politifact.com/>.

Table 1
Summary of unimodal fake news datasets.

Dataset	Description	Modality	Source	Labels	Data volume
2CTweets [34]	Tweets classification for incident-related events	Text	Twitter	Related, non-related	Not specified
BuzzFeed [35]	Hyperpartisan and fake news articles	Text	Facebook	Mostly true, mixture, mostly false, no factual content	1627
COVID-19 [36]	COVID-19 fake news dataset	Text	Public fact-verification sites, Social media	Real (22,916), Fake (19,728)	42,644
CREDBANK [32]	Social media credibility annotations corpus	Text	Global tweets	Credibility annotations	1000
CrisisLex [37]	Crisis-related lexicon for microblogs	Text	Twitter	Disaster-related (keyword-based: 9,777,964, location-base: 639,667)	10,417,631
Emergent [38]	Stance classification dataset for rumors	Text	Rumor sites, Twitter	For (1238), Against (394), Observing (963)	2595
FacebookHoax [33]	Hoaxes and non-hoaxes on social networks	Text	Facebook	Hoaxes (8923), Non-hoaxes (6577)	15,500
George McIntire Dataset [39]	Political fake news detection dataset	Text	Various news websites	Fake (3164), Real (3171)	6335
ISOT [31]	Benchmark dataset for fake news detection	Text	Reuters, Politifact, Kaggle	Fake (12,600), Real (12,600)	25,200
LIAR [23]	Truthfulness annotations on short statements	Text	PolitiFact	Pants-fire, False, Barely-true, Half-true, Mostly-true, True	12,836
MICC [40]	Image forensic dataset for copy-move detection	Image	Columbia Photographic Image Repository	Fake, Real	MICC-F220, MICC-F2000, MICC-F6000
PHEME [41]	Rumor detection dataset for major social events	Text	Twitter	Rumor (1972), Non-rumor (3830)	5802
Twitter Indian Dataset v3 [42]	Fake news detection in political and religious content	Image	Twitter	Fake (61), Valid (49)	110

While text-based analysis is essential, it often cannot account for visual deceptions that reinforce false narratives. Consequently, image-based datasets have been developed to analyze these visual cues. These datasets consist of manipulated or misleading visuals paired with annotations, allowing researchers to analyze visual cues such as inconsistencies, tampering, or contextual mismatches. Unimodal datasets not only aid in the development of modality-specific detection methods but also serve as a foundation for more sophisticated multimodal approaches by providing a deeper understanding of the unique attributes of each modality.

The MICC [40] dataset is a forensic images designed for the detection and analysis of copy-move forgeries. It consists of three subsets, each tailored to different scales and complexities in forgery detection. MICC-F220 contains 220 images with copy-move forgeries, MICC-F2000 includes 2000 images, and MICC-F6000 is a comprehensive set of 6000 high-resolution images with copy-move forgeries. This dataset is effective for learning the characteristics of image manipulation and evaluating the performance of algorithms designed for forgery detection. The MICC dataset is particularly suitable for testing traditional methods, such as SIFT-based clustering, and advanced deep learning techniques for detecting synthetic manipulations. Its diverse subsets make it a valuable resource for developing robust models capable of identifying and localizing image forgeries across various scenarios.

The Twitter Indian Dataset v3 [42] focuses on analyzing fake and authentic images shared on Twitter in India between November 2019 and November 2020, with a particular emphasis on political and religious events, which are common targets of fake news. The dataset consists of 110 images, of which 61 are labeled as fake and 49 as authentic. These images were collected by searching for morphed or forged news images on Indian fact-checking websites such as BoomLive, Alt News, and India Today, and then retrieving the corresponding tweets and news articles from Twitter. The dataset underwent a two-phase verification process: initially, images were cross-checked on fact-checking websites, and subsequently, they were manually annotated by peer reviewers who confirmed the authenticity of the images on the Twitter platform. This dataset provides valuable insights for studying the spread of fake images on social media and developing models aimed at detecting fake news.

3.2. Multimodal-based dataset

Unimodal datasets are essential for understanding the distinct characteristics of fake news within each modality. However, in real-world scenarios, fake news often involves multiple modalities such as text paired with images or videos. To address this complexity, multimodal datasets emerged as crucial resources, enabling the integration of diverse modalities to capture the rich and nuanced nature of fake news. By combining text, images, and videos with synchronized annotations, these datasets facilitate the analysis of cross-modal relationships and offer a more realistic representation of the fake news. This supports the development of advanced

J. Lee 等 人

表 1单模态虚假新闻数据集 汇总 。

D数据集	描述 iption	M模态	来源	标签	D ata 数据量
2CTw eets [34]	Tweets 分类 f or 事件相关 事件	Text	Twitte r	相关, 非相关 ated	N 未 指定
BuzzF eed [35]	Hyperpart isa n 和 虚假 新闻 文章	Text	Facebook	Mostly y 真实, 混合, m ostly false, no f actua l 内容	1627
COVID-19 [36]	COVID-19 虚假 新闻 数据集	Text	公共 事实核查 网站 , 社交 media	Real (22,916), Fak e (19,728)	42,644
CREDBANK [32]	Social m 媒体 可信 度 ty 标注 语料库	Text	Global 推文	可信度 标 nn注	1000
CrisisLex [37]	与危机相关的 词汇 n 用于 微博	Text	Twitte r	Disaste r-rel ated (关键词 : 9,777,964, location-b ase: 639,667)	10,417,63 1
Em 涌现 [38]	立场 c lassification 数据集 针对 谣言	Text	Rumor sites, Twitter	对于 (12 38), Again st (394), 观察 g (963)	2595
FacebookH 骗局 [33]	恶作剧 与 非骗 局 观 n 社交 网络 s 政治 虚假 n 闻 检测 n 数据集	Text	Facebook	H 骗局 (8923) , Non-h 恶作剧 (6577)	15,500
George McIntire 数据集 [39]	数据集 fo r 虚假	Text	各类 新闻 websites	Fak e (3164), 实 l (3171)	6335
ISOT [31]	基准 新闻 detectio n	Text	路透社 , PolitiFact , Kaggle	虚假 (12 600) , 雷 l (12,600)	25,200
LIAR [23]	真实性 a nnotati ons o n 简短 声明	Text	PolitiFact	裤子着火 e, False, Barely-true, 半真半假 ue, M 基本属实, True	12,836
MICC [40]	图像 取证 数据集 fo r 复制-移动 检测	Image	Columbi a 摄影 图像 仓库	Fak e, 真实	MICC-F220, MICC-F2000, MICC-F6000
PHEME [41]	Rumor detectio n 数据集 fo r 主要 社会 l 事件	Text	Twitter	谣言 (1972), 非谣言 (3830)	5802
Twitte r 印度 D 数据集 v3 [42]	Fak e 新闻 检测 n in 政治 与 宗教 内容	Image	Twitter	F 虚假 (61), 有效 (49)	110

虽然基于文本的分析至关重要，但它通常无法解释强化虚假叙事的视觉欺骗。因此，基于图像的数据集被开发出来以分析这些视觉线索。这些数据集由经过篡改或具有误导性的视觉内容以及配对的标注组成，使研究人员能够分析诸如不一致性、篡改或语境不匹配等视觉线索。单模态数据集不仅有助于开发特定模态检测方法，而且通过提供对每种模态独特属性的更深入理解，为更复杂的多模态方法奠定了基础。

MICC [40] 数据集是一个专为复制移动伪造的检测和分析而设计的取证图像数据集。它由三个子集组成，每个子集都针对伪造检测中不同的尺度和复杂性而定制。MICC-F220包含220张带有复制移动伪造的图像，MICC-F2000包含2000张图像，而MICC-F6000则是一个包含6000张带有复制移动伪造的高分辨率图像的综合集。该数据集对于学习图像篡改的特征以及评估为伪造检测设计的算法的性能非常有效。MICC数据集特别适合测试传统方法（如基于SIFT的聚类）以及用于检测合成篡改的高级深度学习技术。其多样化的子集使其成为开发能够识别和定位各种场景下图像伪造的鲁棒模型的宝贵资源。

Twitter印度数据集v3 [42] 专注于分析2019年11月至2020年11月期间在印度Twitter上分享的虚假图像和真实图像，特别强调政治和宗教事件，这些是假新闻的常见目标。该数据集由110张图像组成，其中61张被标记为虚假，49张被标记为真实。这些图像是通过在印度事实核查网站（如BoomLive、Alt News和India Today）上搜索变形或伪造新闻图像，然后从Twitter检索相应的推文和新闻文章而收集的。该数据集经历了两阶段的验证过程：首先，在事实核查网站上对图像进行交叉核对；随后，由同行评审员进行手动标注，确认图像在Twitter平台上的真实性。该数据集为研究虚假图像在社交媒体上的传播以及开发旨在检测假新闻的模型提供了宝贵的见解。

3.2. 基于多模态的 数据集

单模态数据集对于理解假新闻在每种模态中的独特特征至关重要。然而，在现实场景中，假新闻通常涉及多种模态，例如文本与图像或视频的配对。为了应对这种复杂性，多模态数据集作为重要资源应运而生，使得能够整合各种模态以捕捉假新闻丰富且微妙的本质。通过将文本、图像和视频与同步标注相结合，这些数据集促进了对跨模态关系的分析，并提供了对假新闻更真实的表征。这支持了先进的

detection models that are capable of effectively handling multimodal inputs. Examples of multimodal datasets are presented in [Table 2](#).

The DeepFake Detection Challenge (DFDC) [\[43\]](#) dataset is a large-scale resource developed by Facebook to advance deepfake detection research. The dataset comprises 104,500 manipulated and 23,654 authentic videos. The dataset includes synthetic videos generated using advanced deepfake techniques, featuring facial manipulation and audio alteration. It is widely used to evaluate the performance of video-manipulation-detection algorithms. By providing high-quality realistic synthetic videos, the dataset enables researchers to test their models under practical and challenging conditions. As such, it is a critical resource for developing robust solutions to combat video-based fake news and manipulations.

The AraFacts [\[45\]](#) dataset is a multimodal resource designed to detect fake news and fake news in the Arabic-speaking world. It integrates text, image, and video data, providing a comprehensive foundation for research in the domain. The dataset was constructed using data collected from various fact-checking organizations affiliated with the International Fact-Checking Network (IFCN). It spans several key topics, including politics, health, social issues, religion, and general sciences & arts. The politics category addresses fake news related to elections and political events, whereas the health category focuses on false information concerning medical and public health topics. The social category includes content related to societal issues and cultural narratives, and the religion category includes claims involving religious themes and controversies. Finally, general sciences & arts present broader topics in general knowledge and artistic themes.

The WEIBO [\[57\]](#) dataset includes both text and image data collected from the Chinese social media platform Weibo, categorized into rumors and nonrumors. It consists of 3749 rumor samples and 3783 non-rumor samples collected between May 2012 and November 2018. This dataset examines the interaction between textual and visual elements and explores how images can reinforce rumors related to specific topics. It is an essential resource for fake news detection research in the Chinese context, enabling integrated analyses of text and image modalities within multimodal-learning frameworks.

The MMHS150K [\[51\]](#) dataset is a multimodal collection of text and image data gathered from Twitter, designed to address tasks such as detecting hate speech. It is categorized into several classes, including non-hate, racist, and sexist content, reflecting the diversity of hate speech. MMHS150K provides researchers with an opportunity to study the intricate relationship between language and imagery in the context of harmful content online. Its multimodal nature enhances our understanding of how text and images contribute to the propagation of offensive messages, making it a valuable resource for developing and evaluating more robust detection models for harmful content.

The MediaEval [\[50\]](#) dataset includes both text and visual information related to major disasters, such as Hurricane Sandy (HS) and the Boston Marathon (BM). It comprises 10,757 fake text-image pairs, with 3540 real and 281 fake samples for HS and 460 real and 281 fake samples for BM. This dataset supports research focused on detecting false visual information disseminated through social media during disasters. This is particularly valuable for assessing the reliability of information in emergency contexts and addressing the challenges of fake news in critical situations.

The Multimodal Multilingual Fact-Checked Fake News (MuMIN) [\[55\]](#) dataset is a comprehensive collection of fact-checked, multilingual multimodal data gathered from social media platforms, designed to support fake news detection across a variety of topics, including COVID-19. It incorporates text, images, tweets, and user metadata, covering diverse languages and topics. The key components of the dataset include tweet text with corresponding fact-checking information, visual content linked to the tweets, and metadata about the tweet authors. MuMIN is well-suited for studying the interactions between text and images and analyzing the spread of fake news within specific user groups. This dataset enables a holistic approach to understanding and mitigating fake news in diverse social and linguistic contexts.

The FakeNewsNet [\[48\]](#) dataset is a widely used multimodal resource designed for fake news detection. It includes both text and image data, where the text consists of news article bodies and headlines, and the images are visual content associated with the news articles. In addition to textual and visual data, FakeNewsNet incorporates social context, enabling researchers to study how fake news spreads across social media platforms. The dataset is divided into two categories: political news sourced from PolitiFact and entertainment news from GossipCop. This division allows for a focused analysis of the textual and visual features associated with fake and real news within these specific domains. By integrating multimodal data, FakeNewsNet supports the exploration of correlations between text and images, making it a valuable tool for identifying the unique features of fake news and enhancing the development of detection models.

The Fakeddit [\[47\]](#) dataset is a multimodal resource designed to support fake news detection research and consists of both text and images collected from Reddit. With a total of 1,155,550 samples, including 628,501 fake news samples and 527,049 real news samples, Fakeddit enables sophisticated analyses that combine textual and visual data. This makes it particularly valuable for studies that leverage contextual cues from images to identify fake news. This dataset is widely used to explore textual and visual patterns associated with fake and real news, allowing researchers to evaluate the synergistic effects of multimodal data. Additionally, the integration of Reddit’s unique social context provides an important resource for understanding the creation and dissemination of fake news, offering insights into how fake news propagates within online communities.

The MULTI-Fake-DetectiVE [\[53\]](#) dataset focuses on the Russia-Ukraine war and integrates text and images to support fake news detection and verification. It is structured around two subtasks: Subtask #1 classifies news as certainly fake, probably fake, probably real, or certainly real, whereas Subtask #2 evaluates the intent of posts, categorizing them as misleading, not misleading, or unrelated. This dataset is designed to assess the effectiveness of multimodal approaches, as combining text and images allows researchers to explore complex structures of fake news. MULTI-Fake-DetectiVE serves as a critical resource for developing and evaluating models that analyze the interplay between textual and visual elements in fake news and address the nuanced challenges of fake news detection and intent analysis.

检测 模型 的 发展， 这些模型 能够 有效地 处理 多模态 输入。 多模态 数据集 的 示例 呈现 在 表 2中。

深度伪造检测挑战赛（DFDC）[\[43\]](#) 数据集是Facebook开发的大规模资源，旨在推进深度伪造检测研究。该数据集包含104,500个被操纵的视频和23,654个真实视频。该数据集包括使用先进深度伪造技术生成的合成视频，具有面部操控和音频篡改特征。它被广泛用于评估视频操控检测算法的性能。通过提供高质量、逼真的合成视频，该数据集使研究人员能够在实际且具有挑战性的条件下测试其模型。因此，它是开发强大解决方案以打击基于视频的假新闻和操纵行为的关键资源。

AraFacts [\[45\]](#) 数据集是一个多模态资源，旨在检测阿拉伯语世界的假新闻。它整合了文本、图像和视频数据，为该领域的研究提供了全面的基础。该数据集使用从国际事实核查网络（IFCN）附属的各个事实核查组织收集的数据构建。它涵盖多个关键主题，包括政治、健康、社会问题、宗教以及综合科学与艺术。政治类别涉及与选举和政治事件相关的假新闻，而健康类别则侧重于有关医疗和公共卫生话题的虚假信息。社会类别包括与社会问题和文化叙事相关的内容，宗教类别包括涉及宗教主题和争议的声明。最后，综合科学与艺术呈现了常识和艺术主题的更广泛话题。

WEIBO [\[57\]](#) 数据集包含从中国社交媒体平台微博收集的文本与图像数据，分为谣言和非谣言两类。该数据集由2012年5月至2018年11月期间收集的3749个谣言样本和3783个非谣言样本组成。该数据集考察了文本与视觉元素之间的交互，并探讨了图像如何强化与特定主题相关的谣言。它是中国语境下假新闻检测研究的重要资源，能够在多模态学习框架内实现对文本与图像模态的整合分析。

MMHS150K [\[51\]](#) 数据集是从Twitter收集的文本与图像多模态数据集，旨在完成检测仇恨言论等任务。它被分为多个类别，包括非仇恨、种族主义和性别歧视内容，反映了仇恨言论的多样性。MMHS150K为研究人员提供了在网络有害内容上下文中研究语言与图像之间复杂关系的机会。其多模态特性加深了我们对文本与图像如何促进攻击性信息传播的理解，使其成为开发和评估更稳健的有害内容检测模型的宝贵资源。

MediaEval [\[50\]](#) 数据集包含与重大灾难相关的文本与视觉信息，例如飓风桑迪（HS）和波士顿马拉松（BM）。它包含10,757个虚假文本-图像对，其中HS有3540个真实样本和281个虚假样本，BM有460个真实样本和281个虚假样本。该数据集支持专注于检测灾难期间通过社交媒体传播的假视觉信息的研究。这对于评估紧急上下文中信息的可靠性，以及应对关键情况下的假新闻挑战，具有特别的价值。

多模态多语言事实核查假新闻（MuMIN）[\[55\]](#) 数据集是一个综合性的集合，包含从社交媒体平台收集的经过事实核查的多语言多模态数据，旨在支持包括COVID-19在内的各种主题的假新闻检测。它包含文本、图像、推文和用户元数据，涵盖多种语言和主题。该数据集的关键组件包括带有相应事实核查信息的推文文本、链接到推文的视觉内容，以及关于推文作者的元数据。MuMIN非常适合研究文本与图像之间的交互，并分析假新闻在特定用户群体中的传播。该数据集能够实现对不同社交和语言上下文中假新闻的理解与缓解的整体方法。

FakeNewsNet [\[48\]](#) 数据集是一个广泛用于假新闻检测的多模态资源。它包含文本与图像数据，其中文本由新闻文章正文和标题组成，图像则是与新闻文章相关的视觉内容。除了文本和视觉数据外，FakeNewsNet还融合了社会语境，使研究人员能够研究假新闻如何在社交媒体平台上传播。该数据集分为两类：来自PolitiFact的政治新闻和来自GossipCop的娱乐新闻。这种划分有助于针对这些特定领域中与虚假和真实新闻相关的文本与视觉特征进行重点分析。通过整合多模态数据，FakeNewsNet支持探索文本与图像之间的相关性，使其成为识别假新闻独有特征和促进检测模型发展的重要工具。

Fakeddit [\[47\]](#) 数据集是一个旨在支持假新闻检测研究的多模态资源，由从Reddit收集的文本与图像组成。该数据集共有1,155,550个样本，其中包括628,501个假新闻样本和527,049个真实新闻样本，Fakeddit能够支持结合文本和视觉数据的复杂分析。这使得它对于利用图像中的上下文线索来识别假新闻的研究尤为宝贵。该数据集被广泛用于探索与虚假和真实新闻相关的文本与视觉模式，使研究人员能够评估多模态数据的协同效应。此外，Reddit独特的社会语境的融入，为理解假新闻的创造和传播提供了重要资源，从而深入洞察假新闻如何在网络社区中传播。

MULTI-Fake-DetectiVE [\[53\]](#) 数据集聚焦于俄乌战争，整合了文本与图像以支持假新闻检测与验证。该数据集围绕两个子任务构建：子任务 #1 将新闻分类为确属虚假、可能虚假、可能真实或确属真实，而子任务 #2 评估帖子的意图，将其分类为误导性、非误导性或无关。该数据集旨在评估多模态方法的有效性，因为结合文本与图像使研究人员能够探索假新闻的复杂结构。MULTI-Fake-DetectiVE 作为一项关键资源，用于开发和评估分析假新闻中文本与视觉元素相互作用的模型，并应对假新闻检测与意图分析的细微挑战。

Table 2
Summary of multimodal fake news datasets.

Dataset	Description	Modality	Source	Labels	Data volume
All Data [44]	Multimodal dataset for fake news detection with text and images	Text, Image	Various news articles and images	Real (8074), Fake (11,941)	20,015
AraFacts [45]	Arabic dataset of naturally occurring claims for fact-checking	Text, Image, Video	Fatabyyano, FactuelAFP Arabic, Misbar, Verify-Sy	False, Partly-False, Sarcasm, True	6,222
BIENdata [46]	Dataset for fake news detection from BIEN competition	Text, Image, OCR	News websites	Real (Text: 19,186, Image: 11,064, OCR: 5276), Fake (Text: 19,285, Image: 10,768, OCR: 3286)	68,865
DFDC [43]	DeepFake Detection Challenge dataset for deepfake video detection	Video	Synthesized videos	Real (23,654), Fake (104,500)	128,154
Fakeddit [47]	Multimodal dataset for fine-grained fake news detection on Reddit	Text, Image	Reddit posts	Real (527,049), Fake (628,501)	1,155,550
FakeNewsNet [48]	Repository with news content, social context, and spatiotemporal information	Text, Social context	Fact-checking websites	Real, Fake	Not specified
IFND [49]	Indian dataset for fake news detection with titles, images, and URLs	Text (Title, URL), Image (URL)	Indian news sites	Real (37,809), Fake (7271)	45,080
MediaEval [50]	Dataset focusing on computational verification in multimedia during crises	Text, Image, URL	Twitter	Real (Hurricane Sandy: 3540, Boston Marathon: 460), Fake (Hurricane Sandy: 10,757, Boston Marathon: 281)	15,038
MMHS150K [51]	Hate speech detection in multimodal tweets with images	Text, Image	Twitter	Non-Hate (112,845), Racist (11,925), Sexist (3495), Homophobic (3870), Religion-Based (163), Other (5811)	138,109
MR2 [52]	Multimodal retrieval-augmented rumor detection dataset	Text, Image	Twitter, Weibo	Rumor, Non-Rumor, Unverified Post	14,700
MULTI-Fake-DetectiVE [53]	Multimodal dataset for Russo-Ukrainian war fake news detection	Text, Image	Social media posts, news articles	Real (252), Fake (196), Misleading (485), Not Misleading (705)	1,638
MultiOFF [54]	Multimodal offensive content detection	Text, Image	Social media	Offensive (305), Non-offensive (438)	743
MuMiN [55]	Multilingual multimodal fake news dataset	Text, Tweet, Image, User	Fact-checking organizations, Twitter	Fake news, Factual, Other	984
ReCOVery [29]	Repository for COVID-19 news credibility research	Text, Image	Social media platforms (NewsGuard, Media Bias/Fact Check)	Reliable (1364), Unreliable (665)	2,029
TI-CNN [30]	Dataset for training CNNs on fake news detection	Text, Image	Megan Risda (240 websites), Authoritative news websites	Real (8074), Fake (11,941)	20,015
WEIBO [56]	Chinese dataset for fake news detection with text and images	Text, Image	Weibo posts	Real (Text: 19,186, Image: 20,461), Fake (Text: 19,285, Image: 13,635)	72,567
WEIBO [57]	Dataset for rumor detection using multimodal information	Tweet Text, Image, Social context	Weibo posts	Rumor (3749), Non-Rumor (3783)	7,532

3.3. Domain specific dataset

Moreover, domain-specific datasets are developed to support fake news detection, focusing on particular topics or fields. These datasets facilitate in-depth research into the characteristics and propagation mechanisms of fake news in specific domains. Notably,

表 2多模态虚假新闻数据集 汇总。

D数据集 所有数据 [44]	描述 包含文本和图像的虚假新闻检测多模态数据集	M模态 文本, 图像	来源 各种新闻文章和图像	标签 Real (8074), 虚假 (11,941)	D数据量 20,015
阿拉伯 [45]	事实数据集的自然发生声明用于核查	文本, 图像, 视频	FactuelAFP Arabic, Misbar, Verify-Sy	False, 部分错误, se, 讽刺, True	6,222
BIEN数据 [46]	e新闻检测来自BIEN for fake news competition	文本, 图像, OCR	新闻与websites	Real (文本: 19,186, Image: 11,064, 无字符识别: 5276), Fake (文本: 19,285, Image: 10,768, 无字符识别: 3286)	68,865
DFDC [43]	DeepFake Detection Challenge数据集用于深度伪造视频	视频	合成 videos	Real (23,654), Fake (104,500)	128,154
Fakeddit [47]	多模态数据集 for 细粒度虚假新闻检测在Reddit	文本, 图像	Reddit 帖子	Real (527,049), Fake (628,501)	1,155,550
虚假Fakenewsnet [48]	Repository with news content, social context, and spatiotemporal information	Text, 社交上下文	F事实核查 websites	Real, Fake	N非指定的
IFND [49]	印度数据集 for 虚假新闻 with titles, images, and URLs	文本 (标题, URL), Image (URL)	印度新闻网站	Real (37,809), Fake (7271)	45,080
MediaEval [50]	Dataset focusing on computational verification in multimedia during crises	文本, 图像, URL	Twitter	真实 (飓风 桑迪: 3540, 波士顿马拉松: 460), Fake (飓风 桑迪: 10,757, 波士顿马拉松: 281)	15,038
MMHS150K [51]	Hate speech detection in multimodal tweets with images	文本, 图像	Twitter	非恐同 (112,845), 种族主义 (11,925), 性别歧视 (3495), 恐同 (3870), 基于宗教 (163), 其他 (5811)	138,109
MR2 [52]	多模态检索增强谣言检测数据集	文本, 图像	Twitter, 微博	谣言, 未验证帖子	14,700
MULTI-Fake-DetectiVE [53]	多模态数据集 for 俄乌战争虚假新闻检测	文本, 图像	社交媒体帖子, 新闻文章	Real (252), Fake (196), 误导性 (485), 非误导性 (705)	1,638
MultiOFF [54]	多模态冒犯性内容检测	文本, Image	Social media	Offensive (305), 非冒犯性 (438)	743
MuMiN [55]	Multilingual multimodal fake news dataset	Text, Tweet, Image, 使用 r	Fact-checking organizations, Twitter	虚假新闻, F实际的, 其他	984
ReCOVery [29]	Repository for COVID-19 news credibility research	Text, Image	Social media platforms (NewsGuard, Media Bias/Fact Check)	Reliable (1364), 不可靠 (665)	2,029
TI-CNN [30]	Dataset for training CNNs on fake news detection	Text, 图像	Megan Risda (240 websites), 权威 news 网站	Real (8074), 虚假 (11,941)	20,015
微博 [56]	Chinese dataset for fake news detection with text and images	Text, 图像	Weibo 帖子	Real (文本: 19,186, 图像: 20,461), 虚假 (文本: 19,285, 图像: 13,635)	72,567
WEIBO [57]	Dataset for rumor detection using multimodal information	Tweet Text, Im, 社交语境	Weibo 帖子	Rumor (3749), Non-Rumor (3783)	7,532

3.3. 特定领域数据集

此外, 特定领域数据集被开发以支持假新闻检测, 聚焦于特定主题或领域。这些数据集促进了对特定领域内假新闻特征与传播机制的深入研究。值得注意的是,

datasets centered on the COVID-19 pandemic and the U.S. presidential elections stand out as prominent examples. Such datasets are essential for understanding the unique structures and features of fake news in these areas and are invaluable for developing targeted and effective detection models. By incorporating these field-specific insights, domain-specific datasets enhance overall research effectiveness and expand the scope of fake news detection across diverse and critical real-world environments.

3.3.1. Covid-19

During the COVID-19 pandemic, fake news had a significant social impact worldwide. Datasets related to this crisis focus on analyzing and detecting false information concerning public health and infectious diseases. These resources provide a solid foundation for developing models that can address the challenges of fake news during health crises, ensuring accurate and reliable communication during emergencies.

The ReCOVery [29] dataset is a multimodal resource collected during the COVID-19 pandemic that comprises text, image, and video data. Based on news from reputable sources such as NewsGuard and Media Bias/Fact Check, this dataset offers a robust foundation for fake news research. This facilitates the study of the interplay between multiple modalities to detect and understand COVID-19-related fake news.

The COVID-19 Fake News [36] dataset, in contrast, is a text-based resource gathered from fact-checking platforms, such as PolitiFact, Snopes, and Boomlive. It focuses on analyzing the impact of fake news on public health during the pandemic and is widely utilized for developing models to curb the spread of fake news. This dataset is particularly valuable for examining the linguistic patterns and contextual features of fake news in textual form, supporting the creation of effective tools for detecting fake news in health-related contexts.

3.3.2. US election

The U.S. presidential election, especially the 2016 election, is one of the focal points of fake news research due to the widespread dissemination of fake news through social media platforms. Election-related datasets are instrumental in studying the generation and propagation mechanisms of political fake news articles.

The TI-CNN [30] dataset includes text and image data related to the 2016 U.S. presidential election, specifically designed for fake news detection. This dataset is a valuable resource for analyzing the interplay between textual and visual elements in political fake news and for evaluating multimodal detection models.

The BuzzFeed [35] dataset, which is based on political news collected from Facebook, focuses on understanding how political fake news spreads and gains credibility on social media. This is particularly useful for studying user interactions and reactions to fake news on platforms such as Facebook, providing insights into the behavioral dynamics of fake news dissemination.

The George McIntire [39] dataset includes text data related to the 2016 U.S. presidential election, including 3164 fake and 3171 real news articles. It is widely used to detect political fake news, supporting research on linguistic cues and emotional elements within textual data to better understand the characteristics of fake news in political contexts.

3.4. Taxonomy of misinformation

As multimodal fake news detection matures, the industry shifted from a simple binary of real versus fake to more nuanced labeling schemes. This transition is most evident in the construction of modern benchmarks that categorize misinformation based on intent and the nature of the content’s distortion. As observed by Nakamura et al. [47] and Segura-Bedmar and Alonso-Bartolome [58], fake news can be categorized into several distinct technical and stylistic types, including satire or parody, which has no intention to cause harm but has the potential to fool, and misleading content, which involves the misleading use of information to frame an issue or individual. Other prominent categories include imposter content, where genuine sources are impersonated, false connection, where headlines, visuals, or captions do not support the content, and manipulated content, where genuine information or imagery is manipulated to deceive. Furthermore, some recent frameworks introduced a three-way classification approach. For example, Yang et al. [59] classified news into real, fake, and unverified categories, reflecting the practical ambiguity found in real-time social media streams.

3.5. Discussion on datasets

Multimodal fake news datasets have changed from small-scale manually curated benchmarks to massive automatically collected repositories. This growth introduces a critical trade-off between the data volume and annotation quality. Early datasets, such as LIAR and WEIBO, offer verified labels; however, their limited scale and reliance on simplistic categorical labels often fail to capture the complex spectrum of modern misinformation, including subtle satires or digitally altered contexts. In contrast, massive datasets, including Fakeddit, provide the unprecedented volume required to train deep transformer-based models, but the automated nature of their scraping and labeling processes inevitably introduces noise and potential misalignments between the text and imagery. This quantitative expansion risks compromising the precision of detection systems in high-stakes real-world scenarios, where nuanced judgment is paramount.

Profound domain and cultural biases that restrict the global applicability of current detection frameworks also exist. Most existing benchmarks are heavily centered on US-centric political events or specific global crises, such as the COVID-19 pandemic, which may lead to models that overfit specific linguistic patterns or event-specific cues. This concentration is exacerbated by the predominance of English-language content, which overlooks the unique syntactical, idiomatic, and cultural nuances inherent to

以COVID-19大流行和美国总统选举为中心的数据集是突出的例子。此类数据集对于理解这些领域中假新闻的独特结构和特征至关重要，对于开发有针对性且有效的检测模型具有不可估量的价值。通过纳入这些特定领域的见解，特定领域数据集提高了整体研究效率，并扩展了假新闻检测在不同且关键的现实环境中的范围。

3.3.1. COVID-19

在COVID-19大流行期间，假新闻在全球范围内产生了重大的社交影响。与这场危机相关的数据集侧重于分析和检测有关公共卫生和传染病的假信息。这些资源为开发能够应对健康危机期间假新闻挑战的模型提供了坚实的基础，从而确保在紧急情况下进行准确且可靠的沟通。

ReCOVery [29] 数据集是在COVID-19大流行期间收集的多模态资源，包含文本、图像和视频数据。该数据集基于来自NewsGuard和Media Bias/Fact Check等可靠来源的新闻，为假新闻研究提供了坚实的基础。这有助于研究多种模态之间的相互作用，以检测和理解与COVID-19相关的假新闻。

相比之下，COVID-19 Fake News [36] 数据集是一个基于文本的资源，收集自PolitiFact、Snopes和Boomlive等事实核查平台。它专注于分析疫情期间假新闻对公共卫生的影响，并被广泛用于开发遏制假新闻传播的模型。该数据集对于研究文本形式假新闻的语言模式和上下文特征尤为宝贵，有助于为在健康相关上下文中检测假新闻创建有效的工具。

3.3.2. 美国 选举

美国总统选举，特别是2016年选举，由于假新闻通过社交媒体平台的广泛传播，成为了假新闻研究的焦点之一。与选举相关的数据集有助于研究政治假新闻文章的生成和传播机制。

TI-CNN [30] 数据集包含与2016年美国总统选举相关的文本和图像数据，专门为假新闻检测而设计。该数据集是分析政治假新闻中文本和视觉元素之间相互作用以及评估多模态检测模型的宝贵资源。

BuzzFeed [35] 数据集基于从Facebook收集的政治新闻，侧重于理解政治假新闻如何在社交媒体上传播并获得可信度。这对于研究用户在Facebook等平台上对假新闻的互动和反应特别有用，为假新闻传播的行为动态提供了见解。

George McIntire [39] 数据集包含与2016年美国总统选举相关的文本数据，包括3164篇假新闻文章和3171篇真实新闻文章。它被广泛用于检测政治假新闻，支持对文本数据中语言线索和情感要素的研究，以更好地理解政治上下文中假新闻的特征。

3.4. 错误信息 的 分类

随着多模态假新闻检测的成熟，该领域已从简单的真实与虚假二元分类转向更细致的标签方案。这种转变在基于意图和内容扭曲性质对错误信息进行分类的现代基准构建中最为明显。正如Nakamura等人 [47] 以及Segura-Bedmar和Alonso-Bartolome [58]，所观察到的，假新闻可以分类为几种不同的技术和风格类型，包括讽刺或恶搞，其无意造成伤害但具有欺骗的潜力；以及误导性内容，涉及对信息的误导性使用来构陷问题或个人。其他突出的类别包括冒充内容，即冒充真实来源；虚假关联，即标题、图像或说明不支持内容；以及篡改内容，即篡改真实信息或图像以进行欺骗。此外，一些最近的框架引入了三分法分类方法。例如，Yang等人 [59] 将新闻分为真实、假和未经证实类别，反映了实时社交媒体流中存在的实际模糊性。

3.5. 关于 数据集 的 讨论

多模态假新闻数据集已从规模较小手动策划基准转变为大规模的自动收集存储库。这种增长在数据量与标注质量之间引入了关键的权衡。早期数据集（如LIAR和微博）提供了经过验证的标签；然而，其有限的规模和对简单分类标签的依赖，往往无法捕捉现代错误信息的复杂谱系，包括微妙的讽刺或数字篡改的上下文。相比之下，包括Fakeddit在内的大规模数据集，提供了训练基于深度Transformer的模型所需的空前数据量，但其抓取和标注过程的自动化性质不可避免地引入了噪声以及文本与图像之间潜在的不对齐。这种数量上的扩张可能会损害检测系统在高风险现实场景中的精确率，而在这些场景中，细致入微的判断至关重要。

严重的领域和文化偏见也限制了当前检测框架的全球适用性。大多数现有的基准严重集中在以美国为中心的政治事件或特定的全球危机（如COVID-19大流行）上，这可能导致模型过度拟合特定的语言模式或事件特定线索。这种集中现象因英语内容的主导地位而加剧，从而忽略了非西方语境所固有的独特句法、习语和文化细微差别，这些细微差别存在于

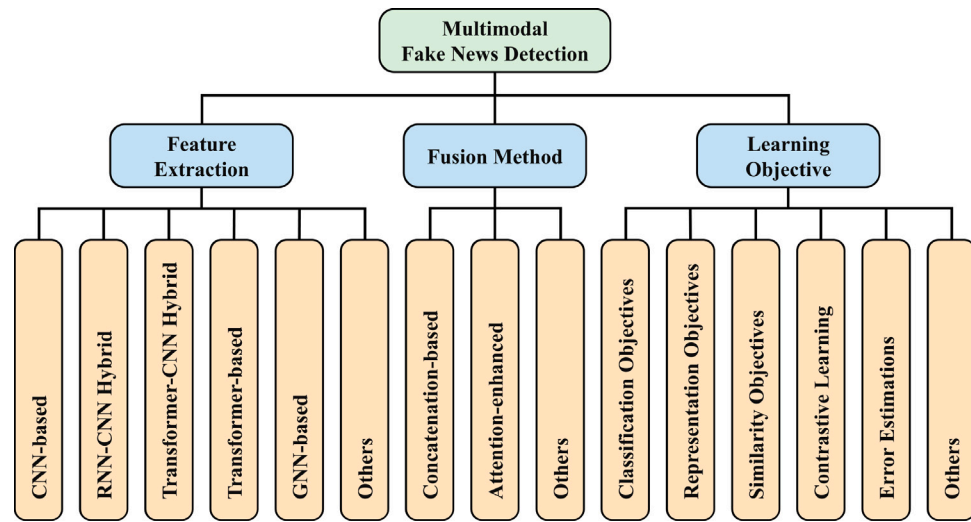


Fig. 4. The structured overview of the model categories. Feature extraction categories are described in Section 4.1, fusion method categories in Section 4.2, and learning objective categories in Section 4.3.

non-Western contexts [60]. Consequently, models validated solely on these benchmarks often struggle with generalizability when they are deployed in diverse linguistic environments. Addressing these biases requires a paradigm shift toward multilingual and culturally diverse data curation, which can only be achieved through collaborative efforts with regional experts and native speakers to ensure that the annotations remain contextually accurate and representative of local disinformation strategies.

The rapid shift in social media consumption highlights a significant modality gap in the current research. While text and image pairs were the primary focus, the rise of short-form video platforms and generative AI necessitates an urgent move toward more complex formats, including audio, video, and interactive synthetic media. Current resources, such as DFDC and AraFacts, begun to address these challenges; however, there remains a substantial need for datasets that provide synchronized, multi-platform, and multi-format data to keep pace with changing deceptive tactics. Ultimately, the research community should prioritize dynamic benchmarking and the integration of emerging AI-generated content to build more robust detection systems that operate reliably across shifting digital landscapes.

4. Multimodal detection

In this section, we analyze multimodal fake news detection models by focusing on three key elements: feature extraction, fusion methods, and the learning objectives. Feature extraction is presented in Section 4.1, fusion methods are discussed in Section 4.2, and the learning objectives are explored in Section 4.3. Figs. 4–6 present a structured overview of the model categories, encoders for each data modality, and fusion methods, respectively. We consider the Web of Science database to select articles. The applied search string is ‘(Machine Learning OR Deep Learning) AND (Multimodal) AND (Fake News) AND (Detect*)’. Among the results, we only consider papers written in English and exclude review papers. The final included papers are summarized in Table 3.

4.1. Feature extraction

The rise of fake news across digital platforms sparks extensive research on multimodal fake news detection. This growing interest is driven by the need to analyze and integrate diverse data modalities, such as text, images, and videos, to accurately identify deceptive content. Multimodal fake news detection harnesses advancements in deep learning and natural language processing (NLP), leveraging state-of-the-art architectures such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), transformer-based models such as BERT [116], and vision transformers (ViTs) [117]. These approaches aim to capitalize on the complementary strengths of various modalities, offering a more holistic perspective on detecting fake news. In this section, we explore the diverse methodologies employed in multimodal models, categorizing them by their architectural designs and highlighting their unique contributions to this rapidly evolving field.

4.1.1. CNN-based models

CNN-based architectures play a crucial role in the early stages of multimodal fake news detection by enabling effective feature extraction from textual and visual data. The majority of these studies proposed frameworks that integrate text-CNNs with pre-trained image CNNs to exploit the complementary nature of these modalities.

One of the pioneering approaches in multimodal fake news detection, Wang et al. [16], introduced the EANN model, which utilized a text-CNN to extract semantic features from textual data and a pre-trained VGG-19 [118] for visual feature extraction. A

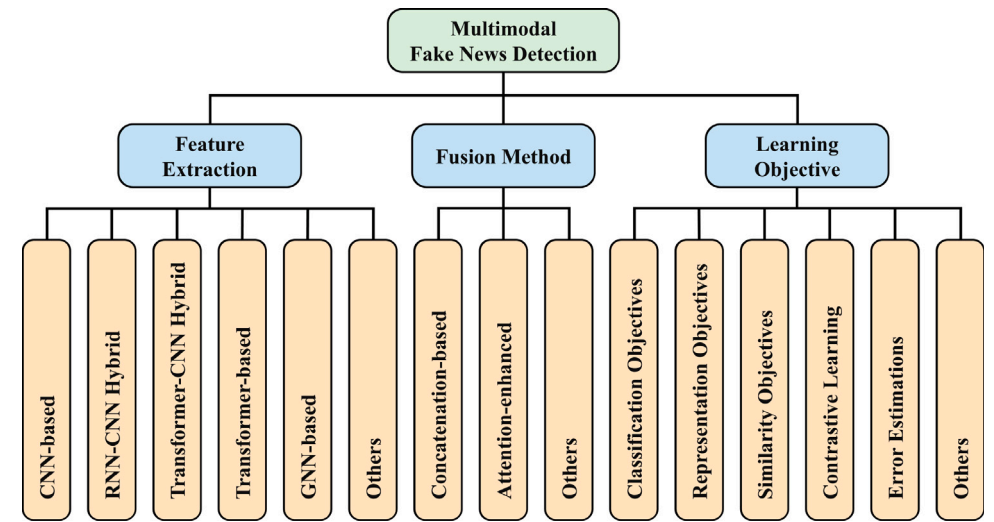


图4. 模型类别的结构化概述。特征提取类别在第4.1节中描述，融合方法类别在第4.2节中，学习目标类别在第4.3节中。

非西方语境[60]中。因此，仅在这些基准上进行验证的模型在部署到多样化的语言环境中时，往往难以具备泛化能力。要解决这些偏见，需要向多语言和文化多样性的数据整理转变，而这只能通过区域专家和母语使用者的合作努力来实现，以确保标注在上下文中保持准确，并能代表当地的虚假信息策略。

社交媒体消费的快速转变突显了当前研究中显著的模态差距。虽然文本和图像对曾是主要焦点，但短视频平台和生成式AI的兴起迫切需要向更复杂的格式转变，包括音频、视频和交互式合成媒体。目前的资源，如DFDC和AraFacts，已开始应对这些挑战；然而，仍然迫切需要提供同步、跨平台和多格式数据的数据集，以跟上不断变化的欺骗手段。最终，研究界应优先考虑动态基准测试和整合新兴的AI生成内容，以构建更强大的检测系统，在不断变化的数字环境中可靠地运行。

4. 多模态 检测

在本节中，我们通过关注三个关键要素来分析多模态假新闻检测模型：特征提取、融合方法和学习目标。特征提取在第4.1节中介绍，融合方法在第4.2节中讨论，学习目标在第4.3节中探讨。图4-6分别展示了模型类别、每种数据模态的编码器以及融合方法的结构化概述。我们考虑使用Web of Science数据库来选择文章。应用的搜索字符串为‘(Machine Learning OR Deep Learning) AND (Multimodal) AND (Fake News) AND (Detect*)’。在结果中，我们仅考虑以英文撰写的论文，并排除综述论文。最终收录的论文总结在表3中。

4.1. 特征 提取

数字平台上假新闻的兴起引发了对多模态假新闻检测的广泛研究。这种日益增长的兴趣源于对分析和整合文本、图像和视频等多种数据模态的需求，以准确识别欺骗性内容。多模态假新闻检测利用了深度学习和自然语言处理（NLP）的进步，采用了最先进的架构，如卷积神经网络（CNN）、循环神经网络（RNN）、基于Transformer的模型（如BERT）[116]，和视觉Transformer（ViT）[117]。这些方法旨在利用各种模态的互补优势，为假新闻检测提供更全面的视角。在本节中，我们探讨了多模态模型中采用的各种方法，根据其架构设计进行分类，并强调它们对这一快速发展领域的独特贡献。

4.1.1. 基于CNN的 模型

基于CNN的架构在多模态假新闻检测的早期阶段发挥着至关重要的作用，能够从文本和视觉数据中进行有效的特征提取。这些研究中的大多数提出了将文本CNN与预训练图像CNN相结合的框架，以利用这些模态的互补性。

多模态 虚假 新闻 检测 的先驱 方法 之一， Wang et al. [16]，引入了 EANN 模型，该模型利用 文本CNN 从 文本 数据中提取 语义 特征，并利用 预训练的 VGG-19 [118] 进行 视觉 特征 提取。A

Table 3
Summary of reviewed approaches; T: Text, I: Image, A: Accuracy, P: Precision, R: Recall, F: F1-Score.

Approach	Modality	Datasets	Evaluation metrics
Qian et al. [61]	<i>T, I</i>	WEIBO, Twitter, PHEME	<i>A, P, R, F</i>
Xiong et al. [62]	<i>T, I</i>	WEIBO, GossipCop	<i>A, P, R, F</i>
Qian et al. [63]	<i>T, I</i> , Knowledge	WEIBO, Twitter, PHEME	<i>A, P, R, F</i>
Wei et al. [64]	<i>T, I</i>	WEIBO, Twitter	<i>A, P, R, F</i>
Raj and Meel [65]	<i>T, I</i>	TI-CNN, Emergent, MICC	<i>A, P, R, F</i>
Luqman et al. [66]	<i>T, I</i>	Fakeddit	<i>A, P, R, F</i> , Matthew's correlation coefficient, Odds ratio
Singhal et al. [67]	<i>T, I</i>	Twitter, Facebook	<i>A, P, R, F</i>
Wang et al. [16]	<i>T, I</i>	WEIBO, Twitter	<i>A, P, R, F</i>
Zhang et al. [68]	<i>T, I</i>	WEIBO, Twitter	<i>A, P, R, F</i>
Al-Alshaqi et al. [69]	<i>T, I</i>	ISOT, MediaEval	<i>A, P, R, F</i>
Athira et al. [70]	<i>T, I</i>	FakeNewsNet	<i>A</i>
Segura-Bedmar and Alonso-Bartolome [58]	<i>T, I</i>	Fakeddit	<i>A, P, R, F</i>
Lv et al. [71]	<i>T, I</i>	WEIBO, Twitter	<i>A, P, R, F</i>
Nakamura et al. [47]	<i>T, I</i>	Fakeddit	<i>A</i>
Yang et al. [59]	<i>T, I</i>	WEIBO, Twitter, IKCEST-2023	<i>A, P, R, F</i>
Sharma and Garg [49]	<i>T, I</i>	IFND	<i>A, P, R</i>
Xu et al. [72]	<i>T, I</i>	WEIBO, Twitter	<i>A, P, R, F</i>
Peng et al. [73]	<i>T, I</i>	WEIBO, Twitter	<i>A, P, R, F</i>
Yadav and Gupta [74]	<i>T, I</i>	Twitter, Fakeddit, Jruvika Fake News, Pontes Fake News, and Risdal Fake News Datasets	<i>A, P, R, F</i>
Hua et al. [75]	<i>T, I</i>	ReCOVery	<i>P, R, F</i>
Li et al. [76]	<i>T, I</i> , OCR	BIENdata, ICT and BAAI Dataset	<i>A, P, R</i>
Sharma et al. [77]	<i>T, I</i>	WEIBO, Fakeddit, MediaEval	<i>A, P, R, F</i>
Lu and Ye [78]	<i>T, I</i> , User-side information	WEIBO	<i>A, P, R, F</i>
Huang et al. [79]	<i>T, I</i>	WEIBO, Fakeddit	<i>A, F</i>
Lin et al. [80]	<i>T, I</i>	Fakeddit, GossipCop	<i>A, P, R, F</i>
Liang [81]	<i>T, I</i>	WEIBO, GossipCop	<i>A, P, R, F</i>
Gôlo et al. [82]	<i>T</i> , Linguistic features	Fake.BR, Fact-Checking News, FakeNewsNet, FakeNewsCorpus	<i>F</i> , AUC-ROC
Kumari and Ekbal [83]	<i>T, I</i>	Twitter, WEIBO	<i>A, P, R, F</i>
Yan et al. [84]	<i>T, I</i> , N-HIN	PHEME, GossipCop, MR2	<i>A, P, R, F</i>
Li et al. [85]	<i>T, I</i>	Twitter, WEIBO	<i>A, P, R, F</i>
Guo [86]	<i>T, I</i>	WEIBO, Twitter	<i>A, P, R, F</i>
Al Obaid et al. [87]	<i>T, I</i>	PolitiFact, GossipCop	<i>A, R, P, F</i> , Specificity, G-Mean
Meel and Vishwakarma [44]	<i>T, I</i>	Jruvika Fake News Dataset, All Data, Pontes Fake News Dataset	<i>A, P, R, F</i>
Rani and Shokeen [88]	<i>T, I</i>	PHEME, CrisisLex, ISOT	<i>A, P, R</i>
Guo et al. [89]	<i>T, I</i> , OCR	WEIBO, Twitter	<i>A, P, R, F</i>
Nadeem et al. [90]	<i>T, I</i>	Jruvika Fake News Dataset, TI-CNN, Pontes Fake News Dataset	<i>A, P, R, F</i>
Qu et al. [91]	<i>T, I</i>	GossipCop, PolitiFact	<i>A, R, P, F</i> , TNR
Qu et al. [92]	<i>T, I</i> , Knowledge, Time	GRFN, IFND	<i>A, P, R, F</i> , MCC
Wang et al. [93]	<i>T, I</i>	PolitiFact, GossipCop, Twitter	<i>A, R, P, F</i>
Yadav et al. [94]	<i>T, I</i>	Twitter, Jruvika Fake News, Pontes Fake News, and Risdal Fake News Datasets	<i>A, P, R, F</i> , AUC-ROC
Yan et al. [95]	<i>T, I</i>	ReCOVery, GossipCop, MR2	<i>A, P, R, F</i> , MCC
Ayetiran and Özgöbek [96]	<i>T, I</i> , OCR	PolitiFact, MMHS150K, MultiOFF	<i>A, P, R, F</i>
Nadeem et al. [97]	<i>T, I</i>	PolitiFact, GossipCop	<i>A, P, R, F</i>
Palani et al. [98]	<i>T, I</i>	PolitiFact, GossipCop	<i>A, P, R, F</i>
Wang et al. [99]	<i>T, I</i>	WEIBO, Twitter	<i>A, F</i>
Yu et al. [100]	<i>T, I</i>	WEIBO, Twitter, PolitiFact, GossipCop	<i>A, P, R, F</i>
Wang et al. [101]	<i>T, I</i> , User-side information	WEIBO	<i>A, P, R, F</i> , AUC, AP
Yin et al. [102]	<i>T, I</i>	Twitter, WEIBO	<i>A, P, R, F</i>
Meel and Vishwakarma [103]	<i>T, I</i>	MediaEval, WEIBO	<i>A, P, R, F</i>
Jarrahi and Safari [104]	<i>T</i> , Social context	FakeNewsNet	<i>A, P, R, F</i>
Zhang et al. [105]	<i>T, I</i>	WEIBO, Twitter, Facebook	TP, FP, TN, FN, Gini coefficient
Sciucca et al. [106]	<i>T, I</i>	Fakeddit	<i>A, F</i>

(continued on next page)

表 3综述 方法 摘要; T: 文本, I: 图像, A: 准确率, P: 精确率, R: 召回率, F: F1分数.

A方法	M模态	数据集	Evaluation 指标
Qian 等人 [61]	<i>T, I</i>	Twitter, PHEME	<i>A, P, R, F</i>
Xiong 等人 [62]	<i>T, I</i>	微博, GossipCop	<i>A, P, R, F</i>
Qi 项目调查与 [63]	<i>T, I</i> , Knowledge	WEIBO, Twitter, PHEME	<i>A, P, R, F</i>
Wei 等人 [64]	<i>T, I</i>	WEIBO, Twitter	<i>A, P, R, F</i>
Raj 和 Meel [65]	<i>T, I</i>	TI-CNN, 涌现, MICC	<i>A, P, R, F</i>
Luqman 等 [66]	<i>T, I</i>	Fakeddit	<i>A, P, R, F</i> , Matthew's 相关系数, 优势比
Singhal 等 [67]	<i>T, I</i>	Twitter, Facebook	<i>A, P, R, F</i>
Wang 等人 [16]	<i>T, I</i>	WEIBO, Twitter	<i>A, P, R, F</i>
Zhang 等人 [68]	<i>T, I</i>	WEIBO, Twitter	<i>A, P, R, F</i>
Al-Alshaqi 等人 [69]	<i>T, I</i>	ISOT, MediaEval	<i>A, P, R, F</i>
Athira 等人 [70]	<i>T, I</i>	FakeNewsNet	<i>A</i>
Segura-Bedmar 和 Alonso-Bartolome [58]	<i>T, I</i>	Fakeddit	<i>A, P, R, F</i>
Lv et al. [71]	<i>T, I</i>	微博, Twitter	<i>A, P, R, F</i>
Nakamura 等人 [47]	<i>T, I</i>	Fakeddit	<i>A, P, R, F</i>
Yang 等人 [59]	<i>T, I</i>	Twitter, IKCEST-2023	<i>A, P, R, F</i>
Sharma and Garg [49]	<i>T, I</i>	微博, IFND	<i>A, P, R</i>
Xu 等人 [72]	<i>T, I</i>	微博, Twitter	<i>A, P, R, F</i>
Peng 等人 [73]	<i>T, I</i>	微博, Twitter	<i>A, P, R, F</i>
Yadav 和 Gupta [74]	<i>T, I</i>	Twitter, Fakeddit, Jruvika 虚假 News, Pontes Fake News, 和 Risdal 虚假 News Datasets	<i>A, P, R, F</i>
Hua et al. [75]	<i>T, I</i>	ReCOVery	<i>P, R, F</i>
Li 等人 [76]	<i>T, I</i> , OCR	BIENdata, ICT 和 BAAI Dataset	<i>A, P, R</i>
Sharma 等人 [77]	<i>T, I</i>	WEIBO, Fakeddit, MediaEval	<i>A, P, R, F</i>
Lu 和 Ye [78]	<i>T, I</i> , 用户侧 通知 ation	WEIBO	<i>A, P, R, F</i>
Huang 等人 [79]	<i>T, I</i>	WEIBO, Fakeddit	<i>A, F</i>
Lin et al. [80]	<i>T, I</i>	Fakeddit, GossipCop	<i>A, P, R, F</i>
Liang [81]	<i>T, I</i>	微博 GossipCop	<i>A, P, R, F</i>
Gôlo 等人 [82]	<i>T</i> , Linguistic features	Fake.BR, F 事实核查 新闻, Fakenewsnet, FakeNewsCorpus	<i>F</i> , AUC-ROC
Kumari 和 Ekbal [83]	<i>T, I</i>	Twitter, 微博	<i>A, P, R, F</i>
Yan 等人 [84]	<i>T, I</i> , N-HIN	PHEME, GossipCop, MR2	<i>A, P, R, F</i>
Li et al. [85]	<i>T, I</i>	Twitter, 微博	<i>A, P, R, F</i>
Guo [86]	<i>T, I</i>	微博, Twitter	<i>A, P, R, F</i>
Al Obaid 等人 [87]	<i>T, I</i>	PolitiFact, GossipCop	<i>A, R, P, F</i> , 特异度, -Mean
Meel 和 Vishwakarma [44]	<i>T, I</i>	Jruvika 虚假 News 数据集, All Data, Pontes Fake News Dataset	<i>A, P, R, F</i>
Rani 和 Shokeen [88]	<i>T, I</i>	PHEME, CrisisLex, ISOT	<i>A, P, R</i>
Guo 等人 [89]	<i>T, I</i> , OCR	微博, Twitter	<i>A, P, R, F</i>
Nadeem 等人 [90]	<i>T, I</i>	Jruvika 虚假 News Dataset, TI-CNN, Pontes Fake News Dataset	<i>A, P, R, F</i>
Qu et al. [91]	<i>T, I</i>	GossipCop, PolitiFact	<i>A, R, P, F</i> , TNR
Qu et al. [92]	<i>T, I</i> , Knowledge, Time	GRFN, IFND	<i>A, P, R, F</i> , MCC
Wang 等人 [93]	<i>T, I</i>	PolitiFact, GossipCop, Twitter	<i>A, R, P, F</i>
Yadav 等人 [94]	<i>T, I</i>	Twitter, Jruvika 虚假 News, Pontes Fake News, 和 Risdal 虚假 News Datasets	<i>A, P, R, F</i> , AUC-ROC
Yan 等人 [95]	<i>T, I</i>	ReCOVe r, GossipCop, MR2	<i>A, P, R, F</i> , MCC
Ayetiran 和 Özgöbek [96]	<i>T, I</i> , 光学字符识别	PolitiFact, MMHS150K, MultiOFF	<i>A, P, R, F</i>
Nadeem 等人 [97]	<i>T, I</i>	Politi事实, 谣言 ipCop	<i>A, P, R, F</i>
Palani 等人 [98]	<i>T, I</i>	Politi事实, 谣言 ipCop	<i>A, P, R, F</i>
Wang 等人 [99]	<i>T, I</i>	WEIBO, Twitter	<i>A, F</i>
Yu 等人 [100]	<i>T, I</i>	WEIBO, Twitter, PolitiFact, GossipCop	<i>A, P, R, F</i>
Wang 等人 [101]	<i>T, I</i> , 用户侧 通知 ation	WEIBO, Twitter, PolitiF	<i>A, P, R, F</i>
Yin 等人 [102]	<i>T, I</i>	微博	<i>A, P, R, F</i> , AUC, AP
Meel 和 Vishwakarma [103]	<i>T, I</i>	Twitter r, 微博	<i>A, P, R, F</i>
Jarrahi 和 Safari [104]	<i>T</i> , Social 上下文	MediaEval, 微博	<i>A, P, R, F</i>
Zhang 等人 [105]	<i>T, I</i>	Fakenewsnet	<i>A, P, R, F</i>
Sciucca 等人 [106]	<i>T, I</i>	微博, Twitter, Facebook k	TP, FP, TN, FN, 基尼 coefficient
	<i>T, I</i>	Fakeddit	<i>A, F</i>

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Table 3 (continued).

Approach	Modality	Datasets	Evaluation metrics
Raj and Meel [107]	T, I	CovID, ReCOVery, CoAID, MediaEval	A, P, R, FP , Specificity, Mathew's correction coefficient
Peng et al. [19]	T, I , Audio	Twitter, WEIBO	A, P, R, F
Goyal et al. [108]	T, I , Social context	Twitter	A, P, R, F
Ahuja and Kumar [20]	T, I , Social context	FakeNewsNet, MKLab-ITI	A, F , Sensitivity, Specificity
Kumari et al. [109]	T , Video, Audio	FakeClips, FVCv2.0, FVCv3.0, VAVD	A, P, R, F
Kumari et al. [110]	T, I , Emotion	NovEmoFake	A, F
Azri et al. [111]	T, I , Sentiment	FakeNewsNet, DAT@Z20	A, P, R, F
Kopev et al. [112]	T , Audio, Metadata	CT-FCC-18	MAE, A, F, R
Zhang et al. [113]	T, I , Knowledge	FakeNewsNet, Evons, Fakeddit, Cosmos, MM-Covid, GoodNews, Twitter, PHEME	A, P, R, F
Zhang et al. [114]	T, I , Propagation information	PolitiFact, GossipCop, PHEME	A, P, R, F
Zhang et al. [115]	T, I , Propagation information	PolitiFact, GossipCop, PHEME	A, P, R, F

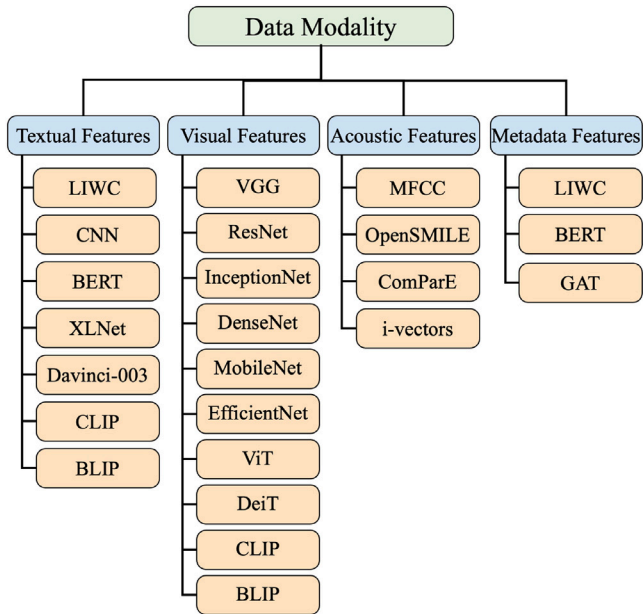


Fig. 5. Encoder of each modality.

distinctive aspect of their approach is the incorporation of adversarial learning, allowing the model to generalize beyond specific news events and effectively identify fake news across diverse contexts. This highlights the potential of CNNs for both text and image modalities and lays the groundwork for integrating adversarial techniques into multimodal architectures.

Building on this foundation, subsequent works explored additional CNN architectures for multimodal fake news detection. For example, Wang et al. [99] employed a similar setup, using a text-CNN for textual features and a pretrained VGG-19 for visual analysis, with an emphasis on effective feature fusion to enhance cross-modal learning. Furthermore, Raj and Meel [65] expanded the scope by evaluating a range of pre-trained CNN models, including ResNet50 [119], VGG-16, InceptionV3 [120], DenseNet [121], and MobileNet [122], combined with a text-CNN for extracting textual features. Their results provide valuable insights into the performance trade-offs between various architectures, offering practical guidance on model selection based on computational resources and dataset characteristics.

Despite the success of CNN-based models in multimodal fake news detection, they exhibit certain limitations, particularly in capturing sequential dependencies within textual data. To address these challenges, several research explored hybrid approaches that combine the sequential modeling capabilities of RNNs with the feature extraction strengths of CNNs. These RNN-CNN-based models aim to bridge this gap by leveraging the complementary strengths of both architectures.

4.1.2. RNN-CNN hybrid models

To address the limitations of CNNs in capturing sequential dependencies, hybrid models that combine RNNs and CNNs are widely adopted for multimodal fake news detection. These RNN-CNN-based models leverage the contextual learning capabilities of RNNs for text data and the feature extraction power of CNNs for images, enabling a more comprehensive multimodal analysis.

表 3 (续)。

Approach	Modality	数据集	Evaluation metrics
Raj 和 Meel [107]	T, I	CovID, ReCOVery, CoAID, MediaEval	A, P, R, FP , 特异性, Mathew's 校正系数
Peng 和 al. [19]	T, I , 音频	Twitter, 微博	A, P, R, F
Goyal et al. [108]	T, I , Social 上下文	Twitter	A, P, R, F
Ahuja 和 Kumar [20]	T, I , 社交 上下文	Fakenewsnet, MKLab-ITI	A, F , Sensitivity, 特异性
Kumari 等人 [109]	T , 视频, 音频	FakeClips, FVCv2.0, FVCv3.0, VAVD	A, P, R, F
Kumari 等人 [110]	T, I , 情感	NovEmoFake	A, F
Azri 等人 [111]	T, I , Sentiment	Fakenewsnet, DAT@Z20	A, P, R, F
Kopev 等人 [112]	T , Audio, Metadata	CT-FCC-18	MAE, A, F, R
张 et al. [113]	T, I , Knowledge	Fakenewsnet, Evons, Fakeddit, GoodNews, Twitter, PHEME	A, P, R, F
张 et al. [114]	T, I , Propagation 信息	PolitiFact, GossipCop, PHEME	A, P, R, F
张 et al. [115]	T, I , Propagation information	PolitiFact, GossipCop, PHEME	A, P, R, F

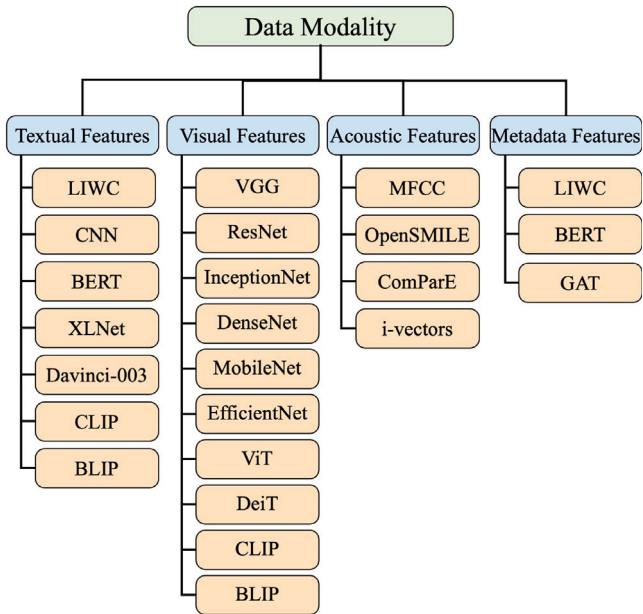


图 5. 各模态的编码器。

他们方法的一个显著特点是引入了对抗学习，使模型能够泛化到特定的新闻事件之外，并在不同的上下文中有效地识别假新闻。这突显了卷积神经网络在文本和图像模态上的潜力，并为将对抗技术整合到多模态架构中奠定了基础。

在此基础上，后续研究探索了用于多模态假新闻检测的其他CNN架构。例如，Wang等人[99]采用了类似的设置，使用文本CNN提取文本特征，并使用预训练VGG-19进行视觉分析，重点在于通过有效的特征融合来增强跨模态学习。此外，Raj和Meel[65]通过评估一系列预训练CNN模型扩大了研究范围，包括ResNet50 [119], VGG-16、InceptionV3 [120], DenseNet[121], 和 MobileNet[122], 并结合文本CNN提取文本特征。他们的结果为各种架构之间的性能权衡提供了宝贵的见解，为基于计算资源和数据集特征进行模型选择提供了实用指导。

尽管基于CNN的模型在多模态假新闻检测中取得了成功，但它们仍表现出一定的局限性，特别是在捕捉文本数据中的序列依赖方面。为了应对这些挑战，一些研究探索了混合方法，将RNN的序列建模能力与CNN的特征提取优势相结合。这些基于RNN-CNN的模型旨在通过利用两种架构的互补优势来弥合这一差距。

4.1.2. RNN-CNN 混合模型

为了解决卷积神经网络在捕捉序列依赖方面的局限性，结合RNN和CNN的混合模型被广泛应用于多模态假新闻检测。这些基于RNN-CNN的模型利用RNN对文本数据的上下文学习能力和CNN对图像的特征提取能力，从而实现更全面的多模态分析。

Kumari and Ekbal [83] introduced an attention-enhanced framework that employed Bi-directional Long Short-Term Memory (Bi-LSTM) [123] with attention mechanisms for textual data, combined with CNN and Bi-directional Gated Recurrent Unit (Bi-GRU) [124] for visual features. Similarly, Raj and Meel [107], Sharma and Garg [49], and Guo [86] used LSTMs to extract sequential text features and pre-trained VGG networks for visual feature extraction. Moreover, Guo [86] demonstrated the importance of low-level visual features such as textures and edges for effective fake news detection. Additionally, Ayetiran and Özgöbek [96] proposed a novel approach that processed not only text and image data but also the relationships between images and text. They utilized tools such as LAVIS [125] for generating image captions and EasyOCR for extracting text embedded in images. Bi-LSTM and CNN layers were then applied to generate feature vectors for these multimodal inputs, demonstrating the potential of integrating textual, visual, and contextual features for enhanced fake news detection.

4.1.3. Transformer-CNN hybrid models

Unlike fully transformer-based models, Transformer-CNN hybrid models retain CNN encoders for visual feature extraction while adopting transformer-based architectures, particularly BERT, for textual encoding. This hybrid design bridges the gap between sequential modeling and contextual understanding, addressing the limitations of RNN-CNN approaches in capturing global context across long textual sequences.

Among Transformer-based models, BERT emerged as a notable innovation. Its bidirectional self-attention mechanism allows for a more effective contextual understanding of the text, addressing the limitations of capturing context and nuanced dependencies across long text sequences. This capability, combined with pre-training on large-scale corpora and fine-tuning for specific downstream tasks, enabled BERT to achieve state-of-the-art performance in various domains, including fake news detection and multimodal analysis.

Following this trend, numerous studies on multimodal fake news detection adopted BERT as a text encoder, leveraging its powerful contextual understanding to enhance performance across both text and image modalities. BERT can be combined with various visual encoders, such as VGG [62,73,93], ResNet [61,69,80], EfficientNet [77], and Inception-ResNet-v2 [103]. Representatively, Nakamura et al. [47] introduced the Fakeddit dataset [126], which evaluated multiple pre-trained text models (BERT, InferSent [127]) and image models (VGG-16, ResNet50, EfficientNet-b4 [128]), providing a valuable foundation for future multimodal studies. However, to overcome the limitation of pooling operations, which ignore spatial relationships in images, Palani et al. [98] employed BERT and CapsNet [129] to extract textual and visual features, respectively. CapsNet’s ability to capture spatial hierarchies in images allowed for a more accurate representation of visual content in the multimodal framework.

Subsequent research [64,71,75] enhanced Transformer-CNN architectures by introducing innovative components. Wei et al. [64] employed BERT and VGG-19 for feature extraction and introduced novel loss functions, such as reconstruction loss, modality discriminator loss, and event discriminator loss, to improve cross-modal representation learning. Hua et al. [75] utilized a back-translation method to augment text data, generating additional text data by translating the original text and then translating it back. They incorporated both the original and back-translated texts, along with the corresponding images, to train the model. Text and image features were extracted using BERT and ResNet50, enabling effective multimodal fake news detection. Furthermore, attention mechanisms within transformers were integrated into hybrid architectures to enhance fake news detection. For example, Guo et al. [89] extracted embedded text from images using PP-OCrV3 [130] and used them as auxiliary features in a multimodal model.

Advancing to BERT-based approaches, several research adopted XLNet [131], which extends BERT using permutation-based training, allowing it to capture bidirectional context more effectively. Al Obaid et al. [87] built an ensemble of attention-based models using XLNet, Bi-GRU, and VGG to aggregate predictions. Qu et al. [91] proposed a novel quantum multimodal fusion-based model, combining quantum encoding and quantum CNNs. They summarized the textual content using text-davinci-003 and embedded it using a pre-trained XLNet. Image features were extracted using VGG. Each modality feature is concatenated and processed using quantum circuits. Amplitude encoding encodes data into quantum states. Three-layer quantum CNNs extract discriminative features from the qubits.

4.1.4. Transformer-based models

In contrast to hybrid approaches aforementioned, fully transformer-based models adopt transformer architectures for both textual and visual modalities. Representative examples include ViT for image patch tokenization and cross-modal pre-trained models such as CLIP and BLIP, which learn joint text-image representations within a unified embedding space.

Huang et al. [79] and Yadav et al. [94] employed ViT for image feature extraction and BERT for textual feature representation, emphasizing the complementary synergy between these two architectures in processing multimodal inputs. Similarly, Yu et al. [100] extended this framework by integrating BERT with Data-efficient Image Transformers (DeiT) [132], a lightweight version of ViT, which enhanced the model’s ability to capture both contextual and visual embeddings with improved computational efficiency.

Building on these architectural advances, recent studies increasingly adopted Contrastive Language-Image Pre-training (CLIP) frameworks [85], marking a significant leap in multimodal learning. In the context of multimodal fake news detection, Xu et al. [72] demonstrated the efficacy of pretrained CLIP models in extracting comprehensive visual-linguistic features. Liang [81] extended this approach by incorporating Bootstrapping Language-Image Pre-training (BLIP) [133], further enhancing multimodal feature extraction. Yan et al. [95] proposed a hybrid framework combining CLIP with BERT and ResNet architectures to optimize both textual and visual feature extraction. Similarly, Zhang et al. [105] implemented an advanced approach combining RoBERTa for textual analysis with CLIP-ViT for visual processing, facilitating robust cross-modal feature fusion. These Transformer-based frameworks represent a significant advancement in multimodal fake news detection, providing robust feature extraction that captures the contextual and semantic nuances of both text and images. However, challenges such as efficient large-scale data processing and effective cross-modal alignment remain, underscoring the need for advanced architectures that can seamlessly integrate multimodal features while maintaining their computational efficiency and cross-modal performance.

Kumari and Ekbal [83] 引入了一个注意力增强框架，该框架采用双向长短期记忆网络（Bi-LSTM） [123] 结合注意力机制处理文本数据，并结合CNN和双向门控循环单元（Bi-GRU） [124] 处理视觉特征。类似地，Raj和Meel [107], Sharma和Garg [49], 以及Guo [86] 使用LSTM提取序列文本特征，并使用预训练的VGG网络进行视觉特征提取。此外，Guo [86] 证明了纹理和边缘等低层视觉特征对于有效假新闻检测的重要性。另外，Ayetiran和Özgöbek [96] 提出了一种新的方法，该方法不仅处理文本和图像数据，还处理图像与文本之间的关系。他们利用LAVIS [125] 等工具生成图像标题，并使用EasyOCR提取嵌入图像中的文本。随后应用Bi-LSTM和CNN层为这些多模态输入生成特征向量，证明了整合文本、视觉和上下文特征以增强假新闻检测的潜力。

4.1.3. Transformer-CNN 混合 模型

与完全基于Transformer的模型不同，Transformer-CNN混合模型保留了CNN编码器用于视觉特征提取，同时采用基于Transformer的架构（特别是BERT）进行文本编码。这种混合设计弥合了序列建模与上下文理解之间的差距，解决了RNN-CNN方法在捕获长文本序列全局上下文方面的局限性。

在基于Transformer的模型中，BERT成为了一项显著的创新。其双向自注意力机制能够更有效地理解文本的上下文，克服了在捕获长文本序列中的上下文和细微依赖关系方面的局限性。这种能力，结合在大规模语料库上的预训练以及针对特定下游任务的微调，使得BERT在包括假新闻检测和多模态分析在内的各个领域都取得了最先进的性能。

顺应这一趋势，许多关于多模态假新闻检测的研究采用BERT作为文本编码器，利用其强大的上下文理解能力来提升文本与图像模态的性能。BERT可以与各种视觉编码器结合使用，例如VGG [62,73,93], ResNet [61,69,80], EfficientNet [77], 和 Inception-ResNet-v2 [103]。具有代表性的是，Nakamura等人 [47] 引入了Fakeddit数据集 [126], , 评估了多种预训练的文本模型（BERT、InferSent [127]）和图像模型（VGG-16、ResNet50、EfficientNet-b4 [128]），为未来的多模态研究提供了宝贵的基础。然而，为了克服池化操作忽略图像中空间关系的局限性，Palani等人 [98] 分别采用BERT和CapsNet [129] 来提取文本与视觉特征。CapsNet捕获图像中空间层次的能力，使得在多模态框架中能够更准确地表示视觉内容。

随后的研究[64,71,75] 通过引入创新组件增强了Transformer-CNN架构。Wei等人[64] 采用BERT和VGG-19进行特征提取，并引入了新颖的损失函数，如重建损失、模态判别器损失和事件判别器损失，以改善跨模态表征学习。Hua等人[75] 利用回译方法来增强文本数据，通过翻译原始文本然后再翻译回来以生成额外的文本数据。他们将原始文本和回译文本连同相应的图像一起用于训练模型。使用BERT和ResNet50提取文本和图像特征，从而实现有效的多模态假新闻检测。此外，Transformer中的注意力机制被集成到混合架构中以增强假新闻检测。例如，Guo等人[89] 使用PP-OCrV3 [130] 从图像中提取嵌入文本，并将其作为多模态模型中的辅助特征。

推进到基于BERT的方法，一些研究采用了XLNet[131], , 它使用基于排列的训练扩展了BERT，使其能够更有效地捕获双向上下文。Al Obaid等人[87] 使用XLNet、Bi-GRU和VGG构建了一个基于注意力的模型集成，以聚合预测结果。Qu等人[91] 提出了一种新颖的基于量子多模态融合 的模型，结合了量子编码和量子卷积神经网络。他们使用text-davinci-003总结了文本内容，并使用预训练的XLNet将其嵌入。使用VGG提取图像特征。每个模态特征被拼接并使用量子电路进行处理。振幅编码将数据编码为量子态。三层量子卷积神经网络从量子比特中提取判别性特征。

4.1.4. 基于Transformer的 模型

与前述的混合方法相比，完全基于Transformer的模型在文本和视觉模态上均采用Transformer架构。代表性示例包括用于图像块分词的ViT，以及如CLIP和BLIP等跨模态预训练模型，这些模型在统一的嵌入空间中学习联合文本图像表征。

Huang等人 [79] 和Yadav等人 [94] 采用ViT进行图像特征提取，并采用BERT进行文本特征表征，强调了这两种架构在处理多模态输入时的互补协同作用。类似地，Yu等人 [100] 通过将BERT与数据高效图像Transformer（DeiT） [132], （ViT的轻量级版本）相结合，扩展了这一框架，从而增强了模型捕获上下文和视觉嵌入的能力，并提高了计算效率。

基于这些架构上的进步，近期的研究越来越多地采用了对比语言-图像预训练（CLIP）框架[85], , 标志着多模态学习的重大飞跃。在多模态假新闻检测的上下文中，Xu等人[72] 证明了预训练的CLIP模型在提取全面的视觉-语言特征方面的有效性。Liang[81] 通过引入自举语言-图像预训练（BLIP） [133], 扩展了这一方法，进一步增强了多模态特征提取。Yan等人[95] 提出了一种将CLIP与BERT和ResNet架构相结合的混合框架，以优化文本和视觉特征提取。同样，Zhang等人[105] 实现了一种结合RoBERTa用于文本分析和CLIP-ViT用于视觉处理的先进方法，促进了稳健的跨模态特征融合。这些基于Transformer的框架代表了多模态假新闻检测的重大进步，提供了稳健的特征提取，能够捕获文本与图像的上下文和语义细微差别。然而，高效的大规模数据处理和有效的跨模态对齐等挑战依然存在，这凸显了对先进架构的需求，这些架构需要能够在保持计算效率和跨模态性能的同时，无缝集成多模态特征。

4.1.5. GNN-based models

The propagation of fake news often occurs through complex social media networks, where the relationships among users, posts, and interactions (e.g., comments, likes, and shares) play a critical role in the dissemination process [134]. Graph Neural Networks (GNNs) are particularly effective for fake news detection because of their ability to model relational data and capture dependencies within graph-structured information. By representing posts, users, and social interactions as nodes and edges in a graph, GNNs provide a comprehensive understanding of the structural and contextual patterns underlying the propagation of fake news. Furthermore, GNNs’ capacity to integrate multimodal and auxiliary information makes them invaluable for addressing the challenges of fake news detection.

Incorporating social context, such as followers and likes, significantly enhances the effectiveness of GNN-based models for detecting fake news. Goyal et al. [108], Yan et al. [84] and Ahuja and Kumar [20] demonstrated the integration of textual, visual, and social features within graph-based frameworks. Goyal et al. [108] processed textual data using LSTMs to extract linguistic and contextual patterns, while visual data, including profile and banner images, were analyzed using CNN architectures. They further incorporated network-based features, such as the number of follows, likes, retweets, and mentions, by encoding these relationships using Graph Convolutional Networks (GCNs) to model the social dynamics of fake news dissemination. Similarly, Ahuja and Kumar [20] expanded the scope by integrating three modalities—textual, visual, and social context—using BERT, VGG, and GNNs [135] to extract and fuse features across modalities. These studies collectively highlight the versatility of GNNs in integrating multimodal and social information to provide a robust framework for detecting fake news. Lastly, Yan et al. [84] constructed a heterogeneous information network that integrated news posts, user profiles, and comments to semantically align textual and visual representations. Their work demonstrated how GNNs effectively capture the relationships between different modalities and social interactions, which are crucial for detecting fake news in complex online environments.

In addition to incorporating social context, GNNs are used to extract features for fake news detection. For example, Qian et al. [63] introduced a knowledge conceptualization module to extract entities from text data. These entities were linked to external knowledge graphs to create conceptual explanations, which, when combined with the text, generated graph-structured data. A Graph Convolutional Network (GCN) [136] was then applied to extract textual features, while image features were processed using a VGG network. Similarly, Qu et al. [92] integrated temporal and content information by clustering news with similar topics and constructing graphs of entities extracted from news texts and images using BERT and a scene graph generator, respectively. Graph Attention Networks (GATs) [137] and GCNs were then employed to generate multimodal and temporal features, emphasizing the importance of leveraging both graph structures and temporal relationships in understanding fake news dissemination patterns.

4.1.6. Others

In multimodal fake news detection, several studies adopted additional features, such as sentiment or emotion, user-specific characteristics, and audio, demonstrating that these features can significantly enhance detection performance. These features offer complementary perspectives to traditional textual and visual modalities, providing nuanced insights into the psychological, social, and auditory dimensions of information dissemination. For example, a series of sentiment and emotion analyses can capture the underlying tone and intent of the content, which can often reveal manipulation or deceptive patterns [138,139]. This section reviews research efforts that incorporate these features, highlighting their contributions to the advancement of multimodal fake news detection systems.

Several approaches were proposed to integrate sentiment and emotion features for fake news detection. In sentiment analysis, rule-based methods such as VADER [111] and TextBlob [66] are commonly employed to detect fake news. Other studies depended on pre-trained deep learning models for more nuanced and robust emotion feature extraction. Wang et al. [101] used sentiment scores derived from SKEP [140], a pre-trained sentiment analysis tool. Kumari et al. [110] categorized images into six emotion classes (joy, love, sadness, fear, surprise, and anger) and extracted emotion features using ResNet. Similarly, Yadav and Gupta [74] extracted emotional features from images using a pre-trained VGG network and from text using emoticons combined with the NRC word-emo

User-oriented features were also utilized in multimodal fake news detection. Lu and Ye [78] extracted text features using BERT and obtained contextual features from posts, such as user profile descriptions and follower/following counts. Several other studies incorporated user-oriented features, including like counts, follower counts, and user activity metrics [101,104].

In addition to textual, visual, and user-oriented features, recent studies explored audio and video modalities for multimodal fake news detection. Kumari et al. [109] and Peng et al. [19] combined BERT for text encoding, VGG-19 for video feature extraction, and various audio processing techniques, including MFCC features and OpenSMILE, to capture acoustic patterns for sentiment analysis. Similarly, Kopev et al. [112] utilized textual features, such as LIWC categories and BERT embeddings, alongside acoustic features derived from ComParE [141] and *i*-vectors [142], integrating metadata such as speaker IDs for a comprehensive multimodal framework.

Recent advancements in multimodal fake news detection attempted to refine feature extraction and fusion techniques by leveraging domain knowledge and integrating diverse modalities into videos. Choi and Ko [143] proposed a domain-adaptive framework combining textual, acoustic, and user-centric features with video-specific cues, incorporating ResNet and BERT-based embeddings to enhance multimodal feature alignment. Furthermore, Qi et al. [144] and Zeng et al. [145] explored a more comprehensive multimodal approach by simultaneously leveraging textual, visual, acoustic, and video modalities. These studies emphasize the importance of incorporating multiple data sources to improve detection accuracy and mitigate the biases inherent in unimodal or weakly multimodal systems. By integrating diverse modalities, they aim to capture the complex nature of fake news spread across social media and video-sharing platforms, making detection more robust and context-aware.

4.1.5. 基于GNN的 模型

假新闻的传播通常通过复杂的社交媒体网络发生，其中用户、帖子和互动（例如，评论、点赞和分享）之间的关系在传播过程中起着至关重要的作用[134]。图神经网络（GNNs）因其建模关系数据和捕获图结构信息内依赖关系的能力，在假新闻检测中尤为有效。通过将帖子、用户和社交互动表示为图中的节点和边，图神经网络为理解假新闻传播背后的结构和上下文模式提供了全面的认识。此外，图神经网络集成多模态和辅助信息的能力使其在应对假新闻检测的挑战时具有不可估量的价值。

结合社会语境（如关注者和点赞）显著提升了基于GNN的模型检测假新闻的有效性。Goyal等人[108]、Yan等人[84]以及Ahuja和Kumar[20]展示了在基于图的框架中融合文本、视觉和社交特征的方法。Goyal等人[108]使用LSTMs处理文本数据以提取语言和上下文模式，而包括头像和横幅图像在内的视觉数据则使用CNN架构进行分析。他们进一步结合了基于网络的特征，如关注数、点赞数、转发数和提及数，通过使用图卷积网络（GCNs）编码这些关系来对假新闻传播的社交动态进行建模。类似地，Ahuja和Kumar[20]通过使用BERT、VGG和图神经网络[135]提取并融合跨模态特征，扩展了整合文本、视觉和社会语境这三种模态的范围。这些研究共同突显了图神经网络在整合多模态和社交信息方面的多功能性，为检测假新闻提供了一个强大的框架。最后，Yan等人[84]构建了一个异构信息网络，整合了新闻帖子、用户画像和评论，以语义对齐文本和视觉表示。他们的研究展示了图神经网络如何有效捕捉不同模态与社交互动之间的关系，这对于在复杂的在线环境中检测假新闻至关重要。

除了结合社会语境外，图神经网络还用于提取假新闻检测的特征。例如，Qian等人[63]引入了知识概念化模块，从文本数据中提取实体。这些实体被链接到外部知识图谱以创建概念解释，当与文本结合时，生成了图结构数据。随后应用图卷积网络（GCN）[136]来提取文本特征，同时使用VGG网络处理图像特征。类似地，Qu等人[92]通过聚类具有相似主题的新闻，并分别使用BERT和场景图生成器从新闻文本和图像中提取实体构建图，从而整合了时序和内容信息。随后采用图注意力网络（GATs）[137]和GCNs来生成多模态和时序特征，强调了利用图结构和时序关系在理解假新闻传播模式中的重要性。

4.1.6. 其他

在多模态假新闻检测中，一些研究采用了额外的特征，如情感、用户特定特征和音频，表明这些特征可以显著提升检测性能。这些特征为传统的文本和视觉模态提供了互补的视角，为信息传播的心理、社交和听觉维度提供了细致的见解。例如，一系列情感分析可以捕捉内容的潜在基调和意图，这通常可以揭示操纵或欺骗模式[138,139]。本节回顾了结合这些特征的研究工作，突出了它们对多模态假新闻检测系统发展的贡献。

人们提出了几种方法来整合情感和情绪特征以进行假新闻检测。在情感分析中，诸如VADER[111]和TextBlob[66]等基于规则的方法常被用于检测假新闻。其他研究则依赖于预训练的深度学习模型，以进行更细致、更鲁棒的情绪特征提取。Wang等人[101]使用了从SKEP[140]，（一种预训练的情感分析工具）导出的情感分数。Kumari等人[110]将图像分为六种情绪类别（喜悦、爱、悲伤、恐惧、惊讶和愤怒），并使用ResNet提取情绪特征。类似地，Yadav和Gupta[74]使用预训练的VGG网络从图像中提取情绪特征，并使用表情符号结合NRC word-emo从文本中提取情绪特征

面向用户的特征也被用于多模态假新闻检测。Lu和Ye[78]使用BERT提取文本特征，并从帖子中获取上下文特征，例如用户个人资料描述和粉丝/关注数。其他几项研究也结合了面向用户的特征，包括点赞数、粉丝数 and 用户活动指标[101,104]。

除了文本、视觉和面向用户的特征外，最近的研究还探索了用于多模态假新闻检测的音频和视频模态。Kumari等人[109]和Peng等人[19]结合了用于文本编码的BERT、用于视频特征提取的VGG-19，以及各种音频处理技术（包括MFCC特征和OpenS MILE），以捕获用于情感分析的声学模式。类似地，Kopev等人[112]利用了诸如LIWC类别和BERT嵌入的文本特征，以及从ComParE[141]和*i*-vector[142]，提取的声学特征，整合了诸如说话人ID的元数据，构建了一个全面的多模态框架。

最近多模态假新闻检测的进展试图通过利用领域知识并将多种模态整合到视频中，来改进特征提取和融合技术。Choi和Ko[143]提出了一个域自适应框架，将文本、声学 and 以用户为中心的特征与视频特定线索相结合，并融入ResNet和基于BERT的嵌入，以增强多模态特征对齐。此外，Qi等人[144]和Zeng等人[145]通过同时利用文本、视觉、声学 and 视频模态，探索了一种更全面的多模态方法。这些研究强调了纳入多种数据源对于提高检测准确率以及减轻单模态或弱多模态系统中固有的偏见的重要性。通过整合多种模态，他们旨在捕捉在社交媒体和视频分享平台上传播的假新闻的复杂性质，使检测更加鲁棒且具备上下文感知能力。

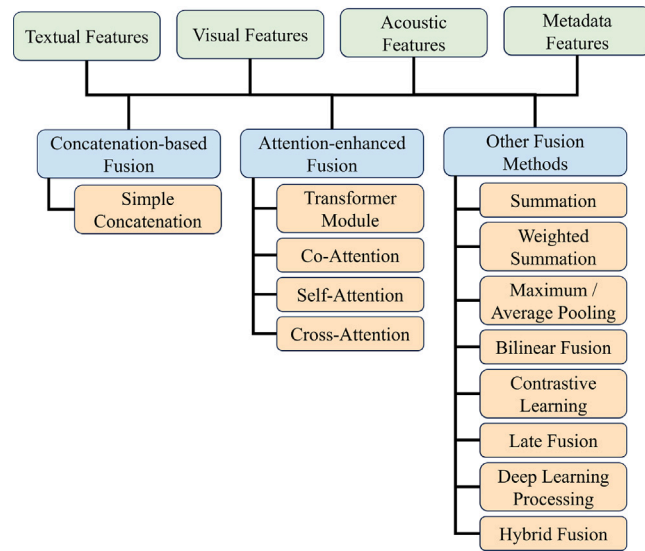


Fig. 6. Diverse fusion methods used to generate multimodal representations explicated in Section 4.2.

In summary, the diverse approaches employed in multimodal fake news detection highlight the potential of leveraging multiple modalities to improve detection performance. CNNs-, RNN-CNN hybrid-, transformer-, and GNNs each demonstrate unique strengths in capturing spatial, temporal, and relational patterns within multimodal data. Moreover, the integration of complementary features, such as sentiment, user-specific metadata, and audio-visual cues, significantly enriches detection frameworks, providing nuanced insights that go beyond traditional text and image analysis. These advancements emphasize the importance of unified and adaptable multimodal approaches for building robust fake news detection systems.

4.1.7. Discussion on feature extraction

Each feature extraction paradigm presents distinct trade-offs in terms of performance, computational cost, and applicability. CNN-based models are lightweight and suitable for real-time scenarios; however, they lack the ability to capture long-range textual dependencies. RNN-CNN hybrids improve sequential text modeling but struggle with global contexts across long documents. Transformer-CNN and fully transformer-based models achieve stronger cross-modal semantic understanding at the cost of higher computational demands, making them less practical in resource-constrained environments than other models. GNN-based models uniquely capture social propagation patterns, offering advantages in detecting coordinated misinformation. However, they require rich social graph data that may not always be available. Overall, the choice of architecture should be guided by the available modalities, computational budget, and the specific nature of the misinformation being targeted.

4.2. Fusion method

4.2.1. Concatenation-based fusion approaches

Several research employed concatenation as a strategy to fuse textual and visual features for multimodal representation. For instance, Wei et al. [64] applied modality and event discriminator losses with feature concatenation to enhance the fusion process. Similarly, Wang et al. [16], Singhal et al. [67], Zhang et al. [68], and Luqman et al. [66] used simple concatenation to integrate image and text features. Segura-Bedmar and Alonso-Bartolome [58] also employed concatenation, but with embeddings extracted via CNN models.

Further studies, such as Sharma and Garg [49] and Sharma et al. [77], and Yadav and Gupta [74], leveraged concatenation to generate robust multimodal features. In addition, Al Obaid et al. [87] and Guo et al. [89] incorporated concatenation with classifier layers to make the final predictions. Kopev et al. [112] significantly highlighted the flexibility of concatenation by integrating features from multiple modalities, including text, image, acoustic, and metadata.

4.2.2. Attention-enhanced fusion techniques

Attention mechanisms, often used in combination with concatenation, are effective for multimodal fusion because they enhance the interaction between textual and visual features. For example, Qian et al. [61] introduced a multimodal contextual attention network that leverages transformer modules to fuse text and image features. Zhang et al. [113] also employed cross attention for the multimodal features. Similarly, Xiong et al. [62] designed an advanced architecture incorporating an Inconsistency Measurement Module that evaluates modality similarity. This is followed by co-attention layers and multiple concatenation operations to achieve

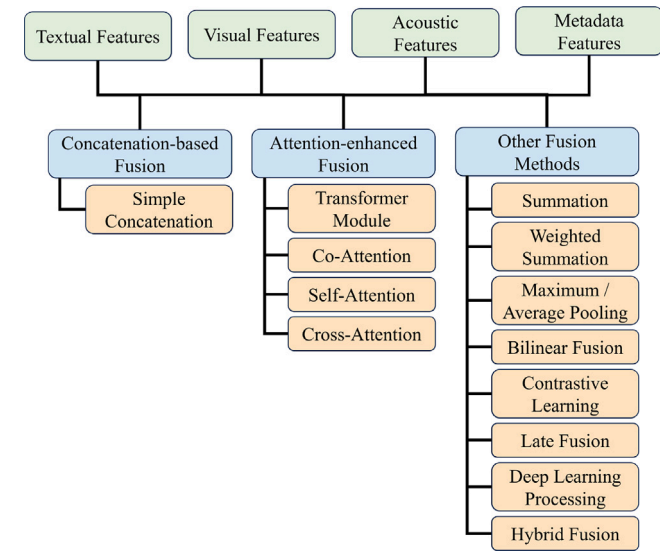


图 6. 第 4.2 节中阐述的用于生成多模态表示的各种融合方法。

综上所述，多模态假新闻检测中采用的多种方法凸显了利用多种模态来提升检测性能的潜力。卷积神经网络、RNN-CNN混合模型、Transformer和图神经网络各自在捕捉多模态数据中的空间、时间和关系模式方面展现出独特的优势。此外，互补特征（如情感、用户特定元数据和视听线索）的整合，极大地丰富了检测框架，提供了超越传统文本与图像分析的细微见解。这些进展强调了统一且适应性强的多模态方法对于构建鲁棒的假新闻检测系统的重要性。

4.1.7. 关于特征提取的讨论

每种特征提取范式在性能、计算成本和适用性方面都呈现出不同的权衡。基于CNN的模型轻量且适用于实时场景；然而，它们缺乏捕获长距离文本依赖的能力。RNN-CNN混合模型改进了序列文本建模，但在处理长文档的全局上下文时存在困难。Transformer-CNN和完全基于Transformer的模型以更高的计算需求为代价实现了更强的跨模态语义理解，这使得它们在资源受限环境中不如其他模型实用。基于GNN的模型独特地捕获了社交传播模式，在检测协同虚假信息方面具有优势。然而，它们需要可能并不总是可用的丰富社交图数据。总体而言，架构的选择应取决于可用的模态、计算预算以及所针对的错误信息的具体性质。

4.2. 融合方法

4.2.1. 基于拼接的融合方法

一些研究采用拼接作为融合文本与视觉特征以形成多模态表示的策略。例如，Wei等人[64]应用了模态与事件判别器损失以及特征拼接来增强融合过程。类似地，Wang等人[16]、Singhal等人[67]、Zhang等人[68]，和Luqman等人[66]使用简单拼接来整合图像与文本特征。Segura-Bedmar和Alonso-Bartolome[58]也采用了拼接，但使用的是通过CNN模型提取的嵌入。

进一步的研究，如Sharma和Garg[49]、Sharma等人[77]，以及Yadav和Gupta[74]，利用拼接来生成鲁棒的多模态特征。此外，Al Obaid等人[87]和Guo等人[89]将拼接与分类器层结合以做出最终预测。Kopev等人[112]通过整合来自多种模态（包括文本、图像、声学 and 元数据）的特征，显著突出了拼接的灵活性。

4.2.2. 注意力增强的融合技术

注意力机制通常与拼接结合使用，对多模态融合非常有效，因为它们增强了文本与视觉特征之间的交互。例如，Qian等人[61]引入了一个多模态上下文注意力网络，利用transformer模块来融合文本和图像特征。Zhang等人[113]也对多模态特征采用了交叉注意力。类似地，Xiong等人[62]设计了一种先进架构，其中包含一个用于评估模态相似度的不一致性度量模块。随后通过共注意力层和多次拼接操作来实现

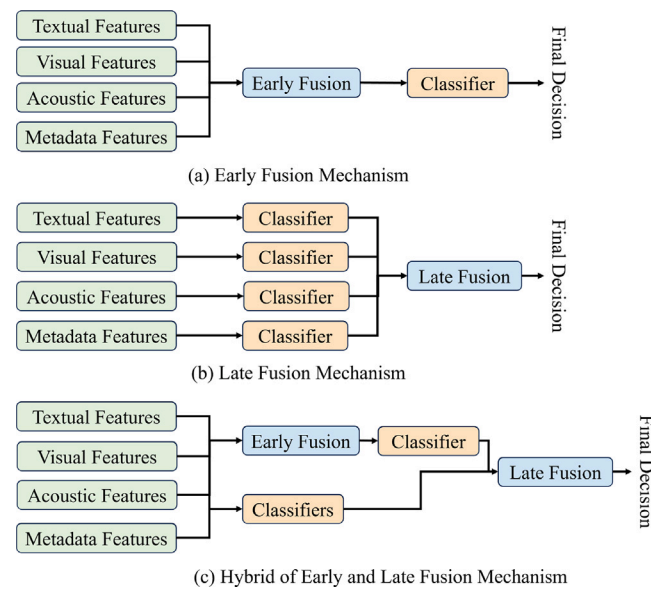


Fig. 7. The different stages of the model in which fusion takes place, discussed in Section 4.2.

a robust multimodal representation. Qian et al. [63] also employed attention mechanisms, applying a co-attention module to GCN and VGG-19 features, followed by pooling and concatenation to finalize feature integration.

In addition, several researchers explored multi-head self-attention for feature fusion. For instance, Hua et al. [75] and Yang et al. [59] utilized self-attention mechanisms to integrate extracted textual and visual features, enhancing modality interactions and generating unified multimodal representations. Peng et al. [73] applied cross multi-head attention to effectively align and fuse text and image embeddings. Similarly, Lu and Ye [78] and Huang et al. [79] leveraged self-attention layers to merge modality-specific features, resulting in improved multimodal fusion performance.

Cross-modal attention emerged as an effective method for feature integration in multimodal fake news detection. Liang [81] employed cross-modal attention to refine modality-specific feature embeddings before their fusion. Similarly, Guo [86] proposed a mutual attention mechanism in which one modality serves as input to the attention mechanism of the other, generating enriched modality features that are then concatenated for the final representation. Further extending these concepts, Yadav et al. [94] introduced visual-semantic and self-attention blocks to extract salient features, ensuring that image and text features complement each other during the fusion process.

Incorporating attention mechanisms into multimodal transformers also yielded promising results. Yan et al. [95] and Zhang et al. [105] utilized co-attention and cross-fusion mechanisms, respectively, to achieve tight alignment between textual and visual modalities. Ayetiran and Özgöbek [96] combined inter-modal attention with weighted feature concatenation, emphasizing the most important features from each modality. Lv et al. [71] further refined attention-based techniques by employing cross-attention on unimodal embeddings, followed by late fusion, to ensure that modality-specific contributions were well represented in the final output. Similarly, Zhang et al. [115] combined self-attention and cross-attention to integrate the temporal and structural features of propagation paths, enhancing the early detection of fake news from multimodal inputs. These studies demonstrate the adaptability of attention mechanisms, enabling deeper and more meaningful interactions between modalities.

4.2.3. Others

Some fusion approaches combine multiple fusion techniques to capture the intricate relationships among modalities. Nakamura et al. [47] explored a variety of fusion methods, including summation, concatenation, maximum pooling, and average pooling, to integrate image and text embeddings, demonstrating the versatility of combining different strategies. Similarly, Xu et al. [72] employed bilinear fusion alongside attention modules to generate robust multimodal features, highlighting the strength of blending simple and advanced fusion techniques for improved feature integration. Zhang et al. [114] combined attention mechanism and diffusion modeling to fuse features of various modalities and constructed propagation graph to detect fake news propagation path.

Contrastive learning is considered a powerful tool in feature fusion frameworks. Kumari et al. [109] applied contrastive learning to align modality-specific features within a shared latent space, ensuring coherence between textual and visual representations. Similarly, Kumari et al. [110] used contrastive learning to align multimodal feature representations with target classes, enhancing inter-modal synergy and class discrimination. These approaches illustrate how these strategies improve feature-level understanding and multimodal representation through the use of alignment techniques. Furthermore, Yan et al. [84] adaptively concatenated contrastively learned, semantically aligned, and network-derived features to create a comprehensive representation of multimodal inputs.

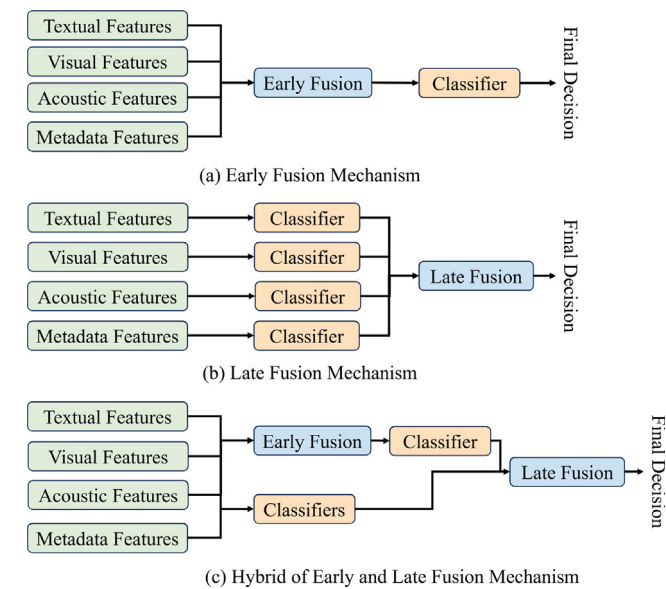


图 7. 模型发生 特征融合 的不同阶段， 在 第 4.2 节中 进行 了 讨论。

鲁棒 的 多模态 表征。 Qian 等人 [63] 也 采用 了 注意力机制， 将 一个 共注意力 模块 应用 于 GCN 和 VGG-19 特征， 随后 通过 池化 和 拼接 来完成 特征 整合。

此外， 一些 研究 人员 探索 了 用于 特征融合 的 多头 自注意力。 例如， Hua 等人 [75] 和 Yang [59] 利用 自注意力 机制 来融合 提取 的 文本 与 视觉 特征， 增强 了 模态 交互 并 生成 了 统一 的 多模态 表示。 Peng 等人 [73] 应用 交叉 多头 注意力 来 有效 对齐 和 融合 文本 和 图像 嵌入。 类似 地， Lu 和 Ye [78] 与 Huang 等人 [79] 利用 自注意力 层 来融合 特定 模态 特征， 从而 提升 了 多模态 融合 的 性能。

跨模态注意力已成为多模态假新闻检测中特征融合的有效方法。 Liang [81] 采用跨模态注意力在融合之前细化特定模态特征嵌入。类似地， Guo [86] 提出了一种互注意力机制， 其中一种模态作为另一种模态注意力机制的输入， 生成丰富的模态特征， 随后将其拼接以形成最终表示。 进一步拓展这些概念， Yadav 等人 [94] 引入了视觉-语义和自注意力块来提取显著特征， 确保图像与文本特征在融合过程中相互补充。

将注意力机制融入多模态Transformer也取得了令人瞩目的成果。 Yan 等人 [95] 和 Zhang 等人 [105] 分别利用共注意力和交叉融合机制， 实现了 文本 与 视觉 模态 之间 的 紧密 对齐。 Ayetiran 和 Özgöbek [96] 将 模态 间 注意力 与 加权 特征 拼接 相结合， 突出 了 每种 模态 中 最重要 的 特征。 Lv 等人 [71] 通过 在 单模态 嵌入 上 采用 交叉 注意力， 随后 进行 后期 融合， 进一步 完善 了 基于 注意力 的 技术， 以 确保 特定 模态 的 贡献 在 最终 输出 中 得到 良好 体现。 同样， Zhang 等人 [115] 结合 自注意力 和 交叉 注意力 来融合 传播 路径 的 时间 和 结构 特征， 从而 增强 了 从 多模态 输入 中 对 假新闻 的 早期 检测。 这些 研究 证明 了 注意力 机制 的 适应性， 能够 实现 模态 之间 更 深层次、 更有 意义 的 交互。

4.2.3. 其他

一些 融合 方法 结合 了 多种 融合 技术， 以 捕捉 模态 之间 错综 复杂 的 关系。 Nakamura 等人 [47] 探索 了 多种 融合 方法， 包括 求和、 拼接、 最大 池化 和 平均 池化， 来融合 图像 和 文本 嵌入， 展示 了 结合 不同 策略 的 通用 性。 同样， Xu 等人 [72] 采用 双线性 融合 与 注意力 模块 相结合 来生成 鲁棒 的 多模态 特征， 突显 了 混合 简单 与 高级 融合 技术 在 改善 特征 融合 方面 的 优势。 Zhang 等人 [114] 结合 注意力 机制 和 扩散 模型 来融合 各种 模态 的 特征， 并构建 了 传播 图 以检测 假新闻 传播 路径。

对比学习被认为是特征融合框架中的强大工具。 Kumari 等人 [109] 应用对比学习在共享潜空间内对齐特定模态特征， 确保了 文本 与 视觉 表示 之间 的 连贯 性。 类似 地， Kumari 等人 [110] 使用 对比 学习 将 多模态 特征 表示 与 目标 类别 对齐， 增强 了 模态 间 协同 和 类别 判别。 这些 方法 说明 了 这些 策略 如何 通过 使用 对齐 技术 来改善 特征 级 理解 和 多模态 表示。 此外， Yan 等人 [84] 自适应 地 拼接 了 对比 学习 的、 语义 对齐 的 和 网络 衍生 特征， 以 创建 多模态 输入 的 综合 表示。

Stacking features and modality-specific processing are key components of advanced fusion strategies. Li et al. [85] stacked learned features and fused them with CNN-MobileNet V2 to create a robust post representation. Azri et al. [111] concatenated text and sentiment features using stacked LSTM layers to create a joint representation, which was then fused with visual features to generate a comprehensive multimodal representation. These studies demonstrate the flexibility and effectiveness of combining stacked architectures with advanced fusion techniques to capture complex multimodal relationships.

Sophisticated fusion method architectures seamlessly integrate early, late, and custom fusion techniques. For instance, Peng et al. [19] introduced a hierarchical approach that combines multiple fusion strategies to capture both intermodal and intramodal correlations. This framework incorporates deep metric learning to enhance features, ensuring robust integration across the text, visual, and acoustic modalities. Similarly, Yin et al. [102] employed a multi-step fusion strategy, beginning with early fusion for basic feature alignment, advancing to joint fusion using co-attention transformers, and concluding with late fusion to integrate the enhanced features. These iterative fusion techniques enable a comprehensive understanding of multimodal inputs.

Other advanced techniques involve the weighted combination of different modalities. Yu et al. [100] combined linear adaptation, self-attention, and gate components to fuse textual and visual features. They then concatenated the fused textual and visual features, along with their differences, to form the final integrated features. Similarly, Nadeem et al. [97] introduced feature stacking and weighted summation to integrate the unimodal and multimodal predictions. Raj and Meel [107] demonstrated the effectiveness of weighted-average fusion by experimenting with various strategies, concluding that this approach outperforms simpler fusion methods.

In conclusion, advanced fusion methods provide flexibility and robustness, making them well suited for a wide range of modalities and tasks. By combining early, late, and joint fusion techniques with attention mechanisms, contrastive learning, and domain-specific knowledge, these methods achieve state-of-the-art multimodal integration performance.

4.2.4. Stages of feature fusion: Early, late, and hybrid fusions

Feature fusion methods in multimodal fake news detection are typically categorized into three types: *early fusion*, *late fusion*, and *hybrid fusion* [146]. This section reviews representative works associated with each fusion strategy and provides the authors' own analysis of their practical implications, trade-offs, and considerations for real-world deployment.

Early fusion. In early fusion, as illustrated in Fig. 7(a), features from different modalities are combined immediately after the feature extraction. Common fusion operations at this stage include concatenation, attention mechanisms, and other vector-level integration techniques such as multi-head self-attention. The resulting multimodal feature representations are then passed to a classifier for prediction.

Late fusion. Late fusion strategies, as shown in Fig. 7(b), involve applying individual classifiers to each modality and then combining the resulting output probabilities. Raj and Meel [65] proposed a weighted late fusion technique, where modality-specific prediction scores are optimized through grid search to improve accuracy. Goyal et al. [108] similarly adopted a weighted summation method to aggregate unimodal outputs, ensuring that each modality contributes appropriately to the final decision. Jarrahi and Safari [104] explored late fusion between latent text features and publisher-related metadata, demonstrating how domain-specific signals can enhance detection performance.

Hybrid fusion. Hybrid fusion combines both early and late fusion stages, as shown in Fig. 7(c). In this approach, features from multiple modalities are first fused through early fusion and then passed to individual classifiers. The resulting predictions are further aggregated using late fusion. Lin et al. [80] investigated this strategy and found that combining joint and decision-level fusion methods improved multimodal performance across various benchmarks.

4.2.5. Discussion on fusion method

Different fusion strategies offer distinct advantages depending on the task requirements, computational budget, and modality availability. Concatenation-based methods remain attractive because of their simplicity, low computational cost, and ease of implementation. These approaches are particularly suitable for real-time fake news detection systems, in which rapid inference is required and all modalities are consistently available. However, simple concatenation often fails to capture the deeper semantic relationships between textual and visual features, which may limit performance in more complex multimodal scenarios.

Attention-based fusion methods provide stronger cross-modal reasoning by explicitly modeling the interactions between modalities. In particular, self-attention, cross-attention, and co-attention mechanisms enable the model to identify salient features and align semantically relevant information across modalities. These methods are particularly effective for detecting fake news involving misleading captions, manipulated images, or subtle inconsistencies between textual and visual content. Nevertheless, these methods typically require substantially higher computational resources and memory consumption than simple fusion techniques, making them less suitable for low-resource or real-time deployment environments.

Similarly, contrastive learning-based fusion methods have shown strong potential for aligning modality-specific representations within a shared embedding space. By encouraging semantically related text-image pairs to be closer while separating unrelated pairs, contrastive learning improves cross-modal consistency and domain generalization. These methods are particularly useful when the goal is to enhance robustness across datasets collected from different domains or social media platforms. However, contrastive learning often requires large-scale training data, carefully designed positive and negative sample pairs, and additional computational overhead during training.

Based on the reviewed literature, each fusion strategy offers unique strengths and tradeoffs. Early fusion methods, such as simple concatenation, are lightweight and computationally efficient, making them suitable for real-time applications, such as fake news

特征堆叠和模态特定处理是高级融合策略的关键组成部分。Li等人 [85] 堆叠了学习到的特征，并将其与CNN-MobileNet V2融合以创建鲁棒的帖子表示。Azri等人 [111] 使用堆叠LSTM层拼接了文本和情感特征以创建联合表示，然后将其与视觉特征融合以生成全面的多模态表示。这些研究证明了将堆叠架构与高级融合技术相结合以捕获复杂多模态关系的灵活性和有效性。

复杂的融合方法架构无缝集成了早期、后期和自定义融合技术。例如，Peng等人[19] 引入了一种层次化方法，该方法结合了多种融合策略以捕获模态间和模态内相关性。该框架结合了深度度量学习以增强特征，确保了跨文本、视觉和声学模态的稳健集成。类似地，Yin等人[102] 采用了多步融合策略，从用于基本特征对齐的早期融合开始，推进到使用共注意力Transformer的联合融合，最后以集成增强特征的后期融合结束。这些迭代融合技术实现了对多模态输入的全面理解。

其他高级技术涉及不同模态的加权组合。Yu等人[100] 结合了线性适配、自注意力和门控组件来融合文本与视觉特征。然后，他们将融合后的文本与视觉特征及其差异进行拼接，以形成最终的集成特征。类似地，Nadeem等人[97] 引入了特征堆叠和加权求和来整合单模态和多模态预测。Raj和Meel[107] 通过实验各种策略，证明了加权平均融合的有效性，并得出结论：该方法优于更简单的融合方法。

总之，高级融合方法提供了灵活性和鲁棒性，使其非常适合广泛的模态和任务。通过将早期融合、后期融合和联合融合技术与注意力机制、对比学习和领域特定知识相结合，这些方法实现了最先进的多模态集成性能。

4.2.4. 特征融合的阶段：早期、后期和混合融合

多模态假新闻检测中的特征融合方法通常分为三种类型：早期融合、后期融合和混合融合[146]。本节回顾了与每种融合策略相关的代表性工作，并提供了作者自己对其实际意义、权衡以及现实世界部署考虑的分析。

早期融合。在早期融合中，如图7(a)所示，来自不同模态的特征在特征提取后立即进行组合。该阶段常见的融合操作包括拼接、注意力机制以及其他向量级集成技术，例如多头自注意力。然后，将得到的多模态特征表示传递给分类器进行预测。

后期融合。后期融合策略如图7(b)所示，涉及对每个模态应用单独的分类器，然后组合得到的输出概率。Raj和Meel [65] 提出了一种加权后期融合技术，其中特定模态的预测分数通过网格搜索进行优化，以提高准确率。Goyal等人 [108] 同样采用加权求和方法来聚合单模态输出，确保每个模态对最终决策做出适当贡献。Jarrahi和Safari [104] 探索了潜在文本特征和发布者相关元数据之间的后期融合，证明了领域特定信号如何增强检测性能。

混合融合。混合融合结合了早期和后期融合阶段，如图7(c)所示。在这种方法中，来自多个模态的特征首先通过早期融合进行融合，然后传递给单独的分类器。得到的预测结果进一步使用后期融合进行聚合。Lin等人 [80] 研究了这种策略，发现结合联合和决策级融合方法提高了各种基准上的多模态性能。

4.2.5. 关于融合方法的讨论

不同的融合策略根据任务需求、计算预算和模态可用性提供不同的优势。基于拼接的方法因其简单性、低计算成本和易于实现而依然具有吸引力。这些方法特别适用于需要快速推理且所有模态均持续可用的实时假新闻检测系统。然而，简单拼接通常无法捕捉文本与视觉特征之间更深层的语义关系，这可能限制其在更复杂的多模态场景中的性能。

基于注意力的融合方法通过显式地建模模态之间的交互，提供了更强的跨模态推理能力。特别是，自注意力、交叉注意力和共注意力机制使模型能够识别显著特征，并在不同模态间对齐语义相关的信息。这些方法对于检测涉及误导性配文、篡改图像或文本与视觉内容之间细微不一致性的假新闻特别有效。尽管如此，这些方法通常比简单的融合技术需要高得多的计算资源和内存消耗，这使得它们不太适合低资源或实时部署环境。

类似地，基于对比学习的融合方法在共享嵌入空间中对齐模态特定表示方面展现出了强大的潜力。通过促使语义相关的文本-图像对更近，同时分离不相关的对，对比学习提升了跨模态一致性和域泛化能力。当目标在于增强从不同领域或社交媒体平台收集的数据集的鲁棒性时，这些方法尤为有用。然而，对比学习通常需要大规模的训练数据、精心设计的正负样本对，以及训练期间额外的计算开销。

基于所综述的文献，每种融合策略都有其独特的优势和权衡。早期融合方法，例如简单的拼接，轻量级且计算高效，这使其适合于实时应用，例如虚假新闻

detection during live debates. However, although more computationally expensive, complex fusion techniques (e.g., attention-based methods) often deliver superior accuracy owing to their ability to model intricate feature interactions.

Late fusion is particularly advantageous in scenarios in which some modalities may be missing or unreliable. Because classification is performed independently for each modality, late fusion supports the flexible integration of partial or incomplete data, which is an important property in dynamic environments.

Hybrid fusion strategies aim to balance performance and robustness, although they typically involve high computational costs. These methods leverage the strengths of both early and late fusion, offering improved generalization, particularly when domain-specific and contextual features are available.

Despite the promising results reported in the literature, several limitations remain. First, most existing methods assume that all modalities are always available, which is often unrealistic in real-world applications. Second, few studies conduct systematic comparisons among early, late, and hybrid fusion methods under the same experimental settings, making it difficult to identify the most effective strategy for a given application scenario. In addition, fake news detection in general remains challenging under noisy, incomplete, or low-quality multimodal inputs, such as blurred images, short texts, or missing contextual information.

Future research may focus on adaptive fusion frameworks that dynamically assign modality weights according to the reliability and quality of each modality. In addition, there is a growing need for lightweight multimodal fusion methods that support real-time inferences in resource-constrained environments. More robust missing-modality fusion strategies, explainable multimodal architectures, and cross-domain generalization techniques are also likely to become important research directions in the future. Finally, recent advances in large language models and vision-language models suggest that trainable fusion modules combined with pre-trained multimodal encoders may provide a promising direction for improving both robustness and generalization in multimodal fake news detection.

4.3. Learning objective

4.3.1. Classification objectives

To detect fake news, the task can be formulated as a binary classification, where the goal is to classify news as either fake or real. Machine learning approaches, such as Logistic Regression (LR) and support vector machines (SVM), were utilized for this task [80]. For deep learning approaches, the sigmoid function is typically applied at the output layer to map the raw score from the model to a probability $p \in [0, 1]$ [72,76,77]. This formulation is optimized using binary cross-entropy loss [65,97,102], defined as follows:

$$\mathcal{L}_{BCE} = -(y \log(p) + (1 - y) \log(1 - p)), \tag{1}$$

where y represents the ground truth label. In contrast, when the task requires identifying multiple specific categories of misinformation, the softmax function is utilized to generate a probability distribution across K classes [70,81,98]. This approach is paired with the categorical cross-entropy loss [63,71,94,100], which minimizes the divergence between the predicted distribution and the one-hot encoded ground truth:

$$\mathcal{L}_{CE} = - \sum_{i=1}^K y_i \log(\hat{y}_i). \tag{2}$$

4.3.2. Representation objectives

Several approaches were proposed to learn better representations for fake news detection. Wei et al. [64] developed a method that learns representations by reconstructing text and image embeddings while simultaneously classifying news as fake or real. Gôlo et al. [82] introduced a multimodal variational autoencoder that reconstructs text-related metrics, achieving more comprehensive multimodal representations for fake news detection. Another approach leverages multitask learning by incorporating networks that classify event categories of news content [16,68,85,86]. By training the model to minimize event-specific performance, event-related information is excluded, enhancing the model’s ability to detect fake news across different events or domains. Kumari et al. [109] emphasized the role of emotional appeal in the dissemination of fake news, integrating audio-visual and textual emotion recognition to capture subtle cues that enrich the model’s representations for better fake news detection.

4.3.3. Similarity objectives

Measuring the similarity among data is crucial for detecting fake news. Meel and Vishwakarma [44] calculated cosine similarity between image captions and news bodies, arguing that fake news involving mismatched text and contextually irrelevant images can be identified by analyzing their coherence. Similar approaches can be applied to measure the similarity between text, images, and OCR text [89], as well as between news body summaries, images, and news headlines [90]. Cosine similarity is also utilized in loss functions to align representations with identical labels [84]. Moreover, techniques such as triplet learning [19,73,79] and Chebyshev distance [93] are mainly employed as loss functions for representation alignment.

检测 在 现场 辩论期间。然而， 尽管 计算 开销 更大， 复杂的 融合 技术 （例如， 基于注意力的 方法） 通常 能提供 更高的 准确率， 因为 它们 具有 能力 来 建模 复杂的 特征 交互。

后期融合在某些模态可能缺失或不可靠的场景中特别有利。由于分类是针对每个模态独立进行的，后期融合支持对部分或不完整数据的灵活整合，这是动态环境中的一项重要特性。

混合 融合 策略 旨在 平衡 性能 和 鲁棒性， 尽管 它们 通常 涉及 高昂的 计算 成本。 这些 方法 利用了 早期 和 后期 融合 的 优势， 提供 了 更好的 泛化能力， 特别是 当 特定领域的 和 上下文 特征 可用 时。

尽管文献中报告了令人瞩目的成果，但仍存在一些局限性。首先，大多数现有方法假设所有模态始终可用，这在实际应用中往往是不切实际的。其次，很少有研究在相同的实验设置下对早期、后期和混合融合方法进行系统比较，这使得难以确定针对特定应用场景最有效的策略。此外，在有噪声、不完整或低质量的多模态输入（如模糊的图像、短文本或缺失的上下文信息）下，假新闻检测总体上仍然具有挑战性。

未来的研究可能会关注自适应融合框架，该框架根据每个模态的可靠性和质量动态分配模态权重。此外，对轻量级多模态融合方法的需求日益增长，以支持在资源受限环境下的实时推理。更鲁棒的缺失模态融合策略、可解释多模态架构以及跨域泛化技术也可能成为未来重要的研究方向。最后，大语言模型和视觉语言模型的最新进展表明，可训练融合模块结合预训练多模态编码器可能为提高多模态假新闻检测的鲁棒性和泛化能力提供一个有前景的方向。

4.3. 学习 目标

4.3.1. 分类 目标

为了检测假新闻，该任务可以被表述为一个二分类问题，其目标是将新闻分类为虚假或真实。机器学习方法，例如逻辑回归（LR）和支持向量机（SVM），已被用于此任务[80]。对于深度学习方法，通常在输出层应用Sigmoid函数，将模型的原始分数映射为概率 $p \in [0, 1]$ [72,76,77]。该公式使用二元交叉熵损失进行优化[65,97,102]，其定义如下：

$$\mathcal{L}_{BCE} = -(y \log(p) + (1 - y) \log(1 - p)), \tag{1}$$

其中 y 代表真实标签。相反，当任务需要识别多种特定类别的错误信息时，使用Softmax函数来生成跨 K 个类别[70,81,98]的概率分布。此方法与分类交叉熵损失[63,71,94,100]，配对使用，以最小化预测分布与独热编码真实标签之间的散度：

$$\mathcal{L}_{CE} = - \sum_{i=1}^K y_i \log(\hat{y}_i). \tag{2}$$

4.3.2. 表示 目标

已提出多种方法来学习更好的假新闻检测表示。Wei等人[64] 开发了一种方法，通过重建文本和图像嵌入并同时新闻分类为虚假或真实来学习表示。Gôlo et al. [82] 引入了一种多模态变分自编码器，用于重建与文本相关的指标，从而为假新闻检测获得更全面的多模态表示。另一种方法利用多任务学习，结合对新闻内容[16,68,85,86]的事件类别进行分类的网络。通过训练模型以最小化特定事件的性能，排除了与事件相关的信息，从而增强了模型在不同事件或领域间检测假新闻的能力。Kumari等人[109] 强调了情感诉求在假新闻传播中的作用，整合了视听和文本情感识别以捕捉微妙的线索，从而丰富模型的表示，以更好地进行假新闻检测。

4.3.3. 相似度 目标

测量数据间的相似度对于假新闻检测至关重要。Meel和Vishwakarma [44] 计算了图像标题与新闻正文之间的余弦相似度，认为通过分析其连贯性，可以识别出包含不匹配文本和上下文无关图像的假新闻。类似方法也可用于测量文本、图像和OCR文本之间的相似度[89]，以及新闻正文摘要、图像和新闻标题之间的相似度[90]。余弦相似度还被用于损失函数中，以对齐具有相同标签的表示[84]。此外，诸如三元组学习[19,73,79] 和切比雪夫距离[93] 等技术主要作为表示对齐的损失函数被采用。

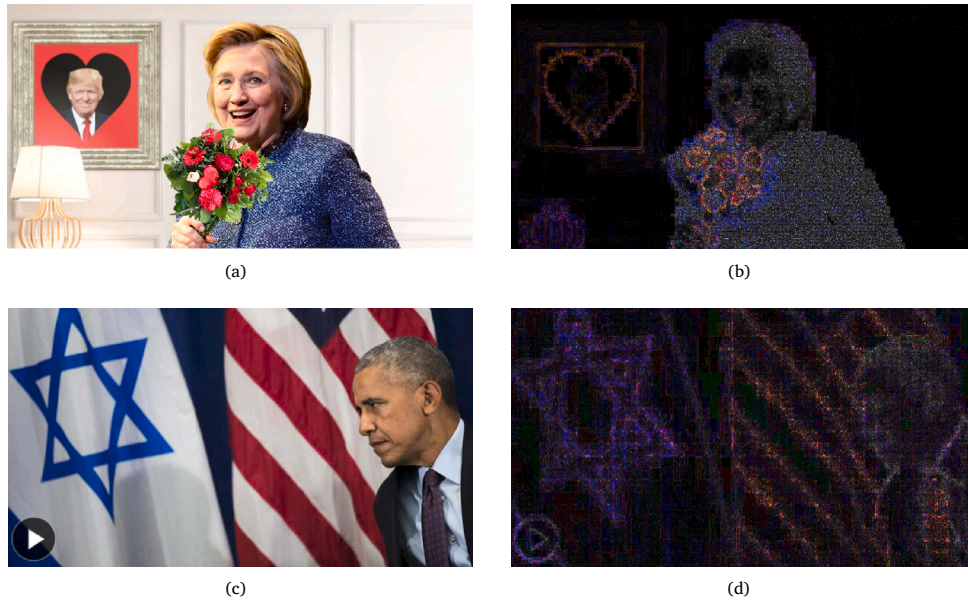


Fig. 8. Examples of ELA. (a) and (c) are forged images, (b) and (d) are ELA-processed images.

4.3.4. Contrastive learning

Contrastive learning is an effective method for generating embeddings of pairs that share similar information while distancing those that have different content. Researchers used contrastive learning to cluster the representations of fake news by utilizing the labels of training samples, ensuring that fake news content is closely represented [19,75,95,110]. Another approach to contrastive learning involves pairing representations from different modalities. Yan et al. [84] paired textual and visual information to position embeddings from the same content close together. Similarly, Kumari et al. [109] employed contrastive learning using text, audio and video embeddings. Beyond modality-level, contrastive learning has also been applied to social graphs to capture structural and semantic relations. For instance, Zhang et al. [115] proposed a dual-view self-supervised contrastive learning framework to jointly model the global propagation topology and high-order heterogeneous relationships among users, publishers, and news content. By integrating node-level and meta-path-based contrastive objectives, their method yields enriched user embeddings, which are subsequently used for reinforced propagation path generation and early-stage fake news detection.

4.3.5. Error estimations

Meel and Vishwakarma [44] estimated noise inconsistencies using K-means clustering on image blocks, asserting that manipulated images exhibit varying noise levels depending on the location within the image. Error Level Analysis (ELA) [147] was employed to assess the extremeness of error levels for fake news detection [44,90]. The ELA is a forensic technique used to determine whether an image was digitally altered. Fig. 8 illustrates examples of ELA.

4.3.6. Others

Wang et al. [99] adopted few-shot learning to enhance fake news detection for emerging events. They utilized meta-learning to efficiently adapt the detector to new events. Al Obaid et al. [87] highlighted the significant class imbalance in real-world news, where real news vastly outnumbers fake news. To address this, they employed focal loss, which is more suitable for imbalanced datasets than cross-entropy loss. In addition, Gôlo et al. [82] used one-class learning and trained their model solely on real news. Qu et al. [92] proposed a two-level cross-entropy loss approach that incorporates temporal information, arguing that consecutive news on the same topic offers complementary insights. Finally, Zhang et al. [105] developed Bayesian classifiers based on Bayes' theorem, leveraging their ability to generalize effectively.

4.3.7. Discussion on learning objective

The evolution of learning objectives in multimodal fake news detection reflects a transition from simple label-based classification to structure-aware representation learning. While classification objectives remain the standard for final decision-making, they often lack the granularity to capture nuanced deceptive patterns between modalities. Binary cross-entropy is computationally efficient but can fail to address the inherent semantic gap between textual and visual data if used in isolation.

In contrast, representation and similarity objectives offer a robust framework by forcing the model to learn the underlying coherence between news bodies and images. These approaches are particularly effective in detecting out-of-context fake news,

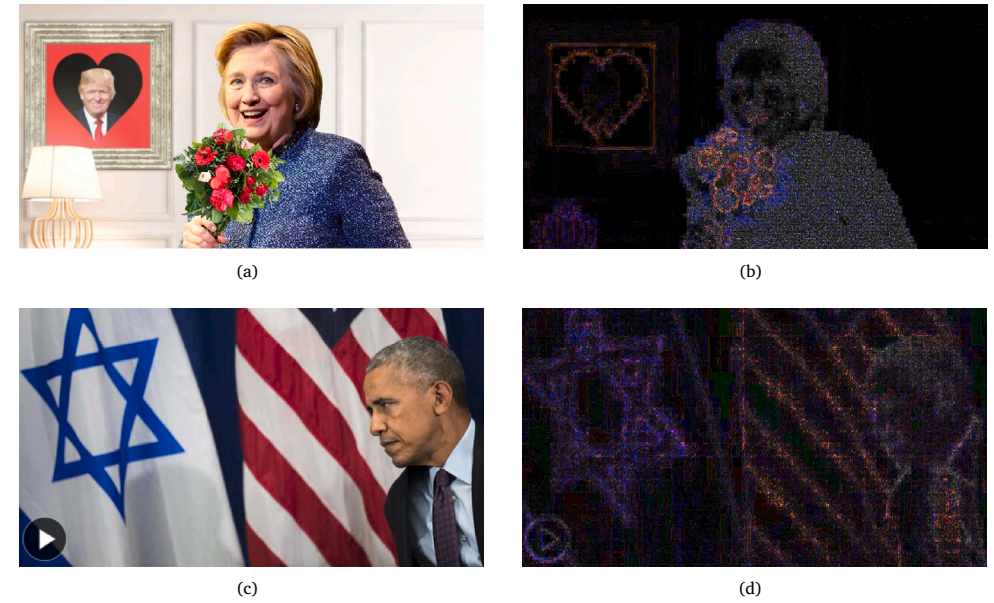


图 8. ELA 示例。(a) 和 (c) 是伪造 图像，(b) 和 (d) 是经ELA处理的 图像。

4.3.4. 对比 学习

对比学习是一种有效的方法，能够为共享相似信息的对生成嵌入，同时使具有不同内容的对相互远离。研究人员利用训练样本的标签，通过对比学习将假新闻的表示进行聚类，确保假新闻内容具有相近的表示[19,75,95,110]。对比学习的另一种方法涉及将来自不同模态的表示进行配对。Yan等人[84]将文本和视觉信息进行配对，使来自相同内容的嵌入位置相近。类似地，Kumari等人[109]使用文本、音频和视频嵌入来进行对比学习。除了模态级别之外，对比学习还被应用于社交图以捕获结构与语义关系。例如，Zhang等人[115]提出了一个双视图自监督对比学习框架，以联合建模全局传播拓扑以及用户、发布者和新闻内容之间的高阶异构关系。通过整合节点级和基于元路径的对比目标，他们的方法产生了丰富的用户嵌入，随后将其用于强化传播路径生成和早期假新闻检测。

4.3.5. 误差 估计

Meel和Vishwakarma[44]利用K均值聚类对图像块估计噪声不一致性，断言被篡改的图像会根据图像内的位置表现出不同的噪声水平。误差水平分析（ELA）[147]被用于评估误差水平的极端程度以进行假新闻检测[44,90]。ELA是一种用于确定图像是否经过数字篡改的取证技术。图8展示了ELA的示例。

4.3.6. 其他

Wang等人[99]采用少样本学习来增强针对新兴事件的假新闻检测。他们利用元学习使检测器能够高效适应新事件。Al Obaid等人[87]强调了现实世界新闻中显著的类别不平衡，即真实新闻的数量远超虚假新闻。为了解决这一问题，他们采用了焦点损失，该损失比交叉熵损失更适合不平衡数据集。此外，Gôlo et al.[82]使用单类学习，仅用真实新闻训练其模型。Qu等人[92]提出了一种结合时序信息的两级交叉熵损失方法，认为同一主题连续新闻能提供互补的见解。最后，Zhang等人[105]基于贝叶斯定理开发了贝叶斯分类器，利用了其有效泛化的能力。

4.3.7. 关于 学习 目标 的讨论

多模态假新闻检测中学习目标的演变，反映了从简单的基于标签的分类到结构感知表示学习的转变。尽管分类目标仍是最终决策的标准，但它们往往缺乏捕捉模态间细微欺骗模式的粒度。二元交叉熵在计算上很高效，但如果单独使用，可能无法解决文本和视觉数据之间固有的语义鸿沟。

相比之下，表征和相似性目标提供了一个稳健的框架，通过迫使模型学习新闻主体和图像之间的潜在连贯性。这些方法在检测脱离上下文的虚假新闻时特别有效，

where the individual modalities are authentic, but their combination is misleading. However, these objectives increase architectural complexity and require carefully curated data to avoid learning trivial correlations.

Contrastive learning aligns high-dimensional features across modalities by maximizing the agreement between related pairs. This enhances the generalizability of the model across different domains and news events. However, the performance of contrastive objectives is highly sensitive to the quality and diversity of negative samples, which poses a challenge for small-scale datasets. Furthermore, as noted in error estimations, learning to identify digital manipulation using forensic-driven objectives adds a critical layer of defense that purely semantic models might overlook.

Finally, addressing real-world data imbalance remains a pivotal challenge. Although focal loss and meta-learning offer potential solutions, their integration into multimodal pipelines can cause training instability. Future research should prioritize the development of hybrid learning objectives that can simultaneously optimize for classification accuracy, cross-modal alignment, and robustness against emerging AI-generated content.

5. Future research

Despite significant advancements in multimodal fake news detection, several challenges persist. By addressing these directions, the field can advance toward creating robust, versatile, and impactful solutions for combating fake news in an ever-evolving digital landscape.

5.1. Cross-platform detection

Most existing studies concentrated on data obtained from a single platform, which limits the generalizability of detection models across different environments. Future research should focus on developing models that can detect fake news across multiple platforms. Each platform presents unique modalities and characteristics, and addressing the absent modalities is key to improving robustness. Approaches such as conditional generative frameworks [148] and cross-modal transformers [149] can be explored to handle incomplete data in multimodal settings, offering a pathway to more adaptable models.

5.2. AI-generated content

As artificial intelligence continues to advance, new challenges in fake news detection are emerging. The evolution of generative models, such as ChatGPT, introduces AI-generated fake news, which exhibits distinct characteristics compared with human-crafted misinformation [150]. These developments underscore the need to create advanced detection models capable of addressing not only traditional fake news but also the unique attributes of AI-generated content. Future research should prioritize the integration of multimodal approaches and leverage cutting-edge techniques to build robust systems that can adapt to this rapidly evolving landscape.

5.3. Modalities

As social media increasingly transitions from text- and image-based content to video-based platforms [14,151], detecting fake news in these new formats becomes imperative. Additionally, with the rise of deepfake technologies, the detection of fake news in video and audio formats is becoming increasingly critical. However, recent research focused primarily on text and image modalities. Future studies should prioritize the development of specialized datasets and models for detecting fake news in video and audio formats to address the emerging challenges and opportunities in this field.

5.4. Contextual information

Modern platforms offer a wealth of auxiliary information beyond the primary content of news items, including user interaction patterns, content metadata, and temporal data. Future research should focus on integrating this auxiliary information into fake news detection models by using appropriate fusion methods. By incorporating such data, models can achieve more nuanced and accurate detection, capturing contextual and relational dynamics that extend beyond the content itself. This approach can provide a deeper understanding of the factors influencing news propagation, ultimately improving the robustness and reliability of the detection systems.

5.5. Learning objectives

Most studies mainly relied on classification-centric objectives, such as cross-entropy loss. However, these objectives may not fully address real-world challenges, including handling imbalanced datasets and capturing uncertainty in predictions. Future research should explore alternative learning objectives. Self-supervised learning approaches, such as bootstrapping [152], can enhance the representation of multimodal data. Adversarial training methods can improve robustness by generating and detecting adversarial examples [153,154]. Adversarial domain adaptation can improve generalizability and adaptability [155]. By diversifying the learning objectives, models can better align with practical requirements and perform more effectively in diverse and dynamic real-world environments.

其中各个模态是真实的，但它们的组合却具有误导性。然而，这些目标增加了架构复杂性，并且需要精心策划的数据以避免学习到琐碎的相关性。

对比学习通过最大化相关对之间的一致性，对齐了跨模态的高维特征。这增强了模型在不同领域和新闻事件中的泛化能力。然而，对比学习目标的性能对负样本的质量和多样性高度敏感，这对小规模数据集构成了挑战。此外，正如误差估计中所指出的，学习使用取证驱动目标来识别数字篡改，增加了一层纯粹的语义模型可能忽略的关键防御。

最后，解决现实世界的​​数据不平衡仍然是一个关键挑战。尽管焦点损失和元学习提供了潜在的解决方案，但将它们整合到多模态流程中可能会导致训练不稳定。未来的研究应优先开发混合学习目标，以同时优化分类准确率、跨模态对齐以及对新兴AI生成内容的鲁棒性。

5. 未来 研究

尽管多模态虚假新闻检测取得了显著进展，但仍有若干挑战持续存在。通过解决这些方向，该领域可以朝着创建稳健、通用且具有影响力的解决方案迈进，以在不断演变的数字环境中打击虚假新闻。

5.1. 跨平台 检测

大多数现有研究集中在从单一平台获取的数据上，这限制了检测模型在不同环境中的泛化能力。未来的研究应侧重于开发能够跨多个平台检测假新闻的模型。每个平台具有独特的模态和特征，解决缺失的模态是提高鲁棒性的关键。可以探索条件生成框架 [148] 和跨模态Transformer [149] 等方法，以处理多模态环境中的不完整数据，为构建更具适应性的模型提供途径。

5.2. AI生成 内容

随着人工智能的不断进步，假新闻检测面临着新的挑战。生成式模型的演进，如ChatGPT，催生了AI生成假新闻，与人工伪造错误信息相比，它展现出截然不同的特征 [150]。这些发展凸显了创建先进检测模型的必要性，此类模型不仅要能够应对传统假新闻，还要能处理AI生成内容的独特属性。未来的研究应优先整合多模态方法，并利用前沿技术来构建鲁棒的系统，以适应这一快速演变的环境。

5.3. 模态

随着社交媒体日益从基于文本和图像的内容向基于视频的平台转型，[14,151]，检测这些新格式中的假新闻变得势在必行。此外，随着深度伪造技术的兴起，检测视频和音频格式的假新闻变得越来越关键。然而，最近的研究主要集中在文本和图像模态上。未来的研究应优先开发专门的数据集和模型，用于检测视频和音频格式的假新闻，以应对该领域新兴的挑战与机遇。

5.4. 上下文 信息

现代平台提供了大量超出新闻项主要内容的辅助信息，包括用户交互模式、内容元数据和时间数据。未来的研究应侧重于通过使用合适的融合方法，将这种辅助信息整合到假新闻检测模型中。通过纳入此类数据，模型可以实现更细致、更准确的检测，捕捉超出内容本身的上下文和关系动态。这种方法可以更深入地理解影响新闻传播的因素，最终提高检测系统的鲁棒性和可靠性。

5.5. 学习 目标

大多数研究主要依赖于以分类为中心的学习目标，例如交叉熵损失。然而，这些学习目标可能无法完全解决现实世界中的挑战，包括处理不平衡数据集和捕捉预测中的不确定性。未来的研究应探索替代的学习目标。自监督学习方法，例如自举法 [152]，，可以增强多模态数据的表示。对抗训练方法可以通过生成和检测对抗样本 [153,154] 来提高鲁棒性。对抗领域自适应可以提高泛化能力和适应性 [155]。通过使学习目标多样化，模型可以更好地符合实际需求，并在多样化和动态的现实世界环境中更有效地运行。

5.6. System-level implementation

Several real-world fact-checking services, such as PolitiFact, address multiple modalities, including images and videos, but they primarily focus on text. These services rely on human-driven verification processes to ensure accuracy and a contextual understanding. However, this human-centric approach inherently limits scalability, making it challenging to keep pace with the growing volume and rapid dissemination of information in the field. This highlights a critical bottleneck in effectively addressing the speed and scope of the propagation of fake news. Consequently, the development and deployment of automated systems are essential to address these challenges and enhance the efficiency and scalability of fact-checking processes. Although most prior research focused on developing detection models, understanding how these models can be effectively deployed in real-world scenarios is equally critical. For instance, Rani and Shokeen [88] proposed a fake news detection system that combines deep learning and blockchain technology to enhance reliability and transparency, highlighting a promising pathway for practical applications. However, research on the practical deployment and integration of detection models into operational systems remains limited. Future studies should explore scalable and efficient system architectures that consider computational resources tailored to diverse contexts. Moreover, addressing ethical concerns, including user privacy and data security, is essential for fostering societal trust and broader adoption, specifically when handling user data such as user metadata or interaction data. Furthermore, such systems must be designed to operate without introducing biases or generating fake news. Transparent mechanisms that provide explainable and interpretable outputs are vital for ensuring accountability and user confidence in such systems.

6. Conclusion

The rapid advancement of digital media and social networks has established fake news as a critical societal challenge that undermines public trust and influences political decisions. While early detection efforts relied on unimodal data, such as text or images, they often fall short of addressing the complex nature of fake news, which frequently combines multiple modalities to create more persuasive and deceptive narratives. In response, multimodal approaches, which integrate and analyze information from various sources, including text, images, videos, and metadata, have emerged as promising and essential solutions.

This study systematically examines the advancements in multimodal fake news detection, providing a comprehensive review of the field and outlining key directions for future research. Specifically, we propose a novel taxonomy that structures multimodal approaches along three critical dimensions: feature extraction, fusion methods, and learning objectives. This framework allows for an in-depth analysis of existing models and a detailed comparison of key datasets, highlighting their characteristics, limitations, and applicability to multimodal tasks.

We trace the evolution of feature extraction from early CNN-based models to more advanced transformer-based architectures (e.g., BERT, ViT) and Graph Neural Networks (GNNs), which better capture contextual and relational data. We analyze the progression of fusion methods from simple concatenation to sophisticated attention-enhanced and hybrid strategies that model complex inter-modal relationships. Finally, we examine the diversity of learning objectives employed, moving beyond standard classification to include representation learning, similarity objectives, and contrastive learning, which enables the construction of more robust models.

This comprehensive analysis reveals several critical challenges, including the lack of diverse datasets that limit model generalizability and the new frontier of detecting AI-generated content, that is, deepfakes. Addressing these hurdles requires future work to focus on integrating auxiliary information, such as user interaction patterns and metadata, to build context-aware systems. Furthermore, exploring alternative learning objectives, such as self-supervised and adversarial training, is key to enhancing model robustness. Ultimately, transitioning these advanced models from research to real-world applications is the final, critical step, requiring scalable, transparent, and ethical systems. This necessitates a collaborative effort between academia, industry, and policymakers to foster a resilient and trustworthy digital-information ecosystem.

CRedit authorship contribution statement

Juyeob Lee: Writing – original draft, Resources, Methodology, Formal analysis, Data curation, Conceptualization. **Junyeop Cha:** Writing – original draft, Supervision, Software, Resources, Investigation, Conceptualization. **Minsu Park:** Writing – original draft, Methodology, Data curation, Conceptualization. **Minyoung Lee:** Writing – original draft, Software, Formal analysis, Data curation, Conceptualization. **Taeun Kim:** Visualization, Supervision, Resources, Funding acquisition, Formal analysis, Data curation. **Seul-Ki Choi:** Visualization, Software, Resources, Funding acquisition, Data curation. **Angel P. del Pobil:** Visualization, Supervision, Funding acquisition. **Eunil Park:** Writing – original draft, Visualization, Validation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

5.6. 系统级 实现

一些现实世界中的事实核查服务，例如PolitiFact，处理包括图像和视频在内的多种模态，但它们主要关注文本。这些服务依赖人工驱动核查过程来确保准确率和上下文理解。然而，这种以人为中心的方法在本质上限制了可扩展性，使其难以跟上该领域不断增长的信息量和信息的快速传播。这凸显了在有效应对假新闻传播速度和范围方面的关键瓶颈。因此，开发和部署自动化系统对于应对这些挑战以及提高事实核查过程的效率和可扩展性至关重要。尽管先前的大多数研究集中在开发检测模型上，但了解如何将这些模型有效地部署到现实世界场景中同样至关重要。例如，Rani和Shokeen [88] 提出了一个结合深度学习和区块链技术的假新闻检测系统，以提高可靠性和透明度，突显了实际应用的一条有前景的途径。然而，关于检测模型在实际运营系统中的部署和集成的研究仍然有限。未来的研究应探索可扩展且高效的系统架构，并考虑为不同上下文量身定制的计算资源。此外，解决包括用户隐私和数据安全在内的伦理问题，对于培养社会信任和更广泛的采用至关重要，特别是在处理用户数据（如用户元数据或交互数据）时。此外，此类系统的设计必须在不引入偏见或生成假新闻的情况下运行。提供可解释和可理解输出的透明机制，对于确保此类系统的问责制和用户信心至关重要。

6. 结论

数字媒体和社交网络的快速发展已使假新闻成为一项严峻的社会挑战，它削弱了公众信任并影响了政治决策。虽然早期的检测工作依赖于单模态数据，如文本或图像，但它们往往无法应对假新闻的复杂性质，因为假新闻经常结合多种模态来制造更具说服力和欺骗性的叙事。作为应对，多模态方法通过整合和分析来自各种来源（包括文本、图像、视频和元数据）的信息，已成为极具前景且必不可少的解决方案。

本研究系统地探讨了多模态假新闻检测的进展，对该领域进行了全面综述，并概述了未来研究的关键方向。具体而言，我们提出了一种新颖的分类法，从三个关键维度构建了多模态方法：特征提取、融合方法和学习目标。该框架允许对现有模型进行深入分析，并对关键数据集进行详细比较，突出了它们的特征、局限性以及对多模态任务的适用性。

我们追溯了特征提取的演变，从早期的基于CNN的模型到更先进的基于Transformer的架构（例如BERT、ViT）和图神经网络（GNNs），这些模型能更好地捕获上下文与关系数据。我们分析了融合方法的演进，从简单拼接到复杂的注意力增强和混合策略，这些方法能够对复杂的模态间关系进行建模。最后，我们探讨了所采用的学习目标的多样性，超越了标准分类，涵盖了表征学习、相似性目标和对比学习，从而能够构建更鲁棒的模型。

这项全面的分析揭示了几个关键挑战，包括缺乏限制模型泛化能力的多样化数据集，以及检测AI生成内容（即深度伪造）的新前沿。要克服这些障碍，未来的工作需要侧重于整合辅助信息（如用户交互模式和元数据），以构建上下文感知系统。此外，探索替代的学习目标（如自监督和对抗训练）是增强模型鲁棒性的关键。最终，将这些先进模型从研究推向现实世界应用是最后也是至关重要的一步，这需要可扩展、透明和伦理的系统。这就需要学术界、产业界和政策制定者之间的通力合作，以培育一个具有韧性且值得信赖的数字信息生态系统。

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Data availability

Data will be made available on request.

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数据 可用性

数据 将 根据 请求 予以 提供 。

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